Linear multidimensional regression with interactive fixed-effects *

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Abstract

This paper studies a linear and additively separable model for multidimensional panel data of three or more dimensions with unobserved interactive fixed effects. Two approaches are considered to account for these unobserved interactive fixed-effects when estimating coefficients on the observed covariates. First, the model is embedded within the standard two-dimensional panel framework and restrictions are formed under which the factor structure methods in Bai (2009) lead to consistent estimation of model parameters, but at slow rates of convergence. The second approach develops a kernel weighted fixed-effects method that is more robust to the multidimensional nature of the problem and can achieve the parametric rate of consistency under certain conditions. Theoretical results and simulations show some benefits to standard two-dimensional panel methods when the structure of the interactive fixed-effect term is known, but also highlight how the kernel weighted method performs well without knowledge of this structure. The methods are implemented to estimate the demand elasticity for beer.

1 Introduction

Models of multidimensional data – panel data with more than two dimensions – are becoming increasingly popular in econometric analysis as large data sets with a multidimensional structure become available. For example, in gravity models of trade that are repeated over time one may be interested in studying trade patterns between an importer, i, an exporter, j, that is repeated

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every quarter or year, t. One may also be interested in studying demand elasticities through consumption data that may vary by product, i, store, j, with repeated observation over week or fortnight, t.[1] In these examples it is clear that there may exist unobserved characteristics in each dimension that can determine variation across both the dependent and independent variables that needs to be controlled for to avoid issues with endogeneity. For example, this could be shifts in taste preferences, that are unobserved by the econometrician, that may effect sales of particular products in certain stores differently over time. Thus far, most analysis has addressed unobserved heterogeneity in the higher-dimensional setting by using a combination of additive scalar fixed-effects. These additive scalar fixed-effects approaches, however, can only accommodate variation in unobserved heterogeneity over a subset of dimensions with any one of the scalar fixed-effects terms. For example, in the three-dimensional model this type of fixed-effect approach can only control for variation over ij, it and jt, but not over all ijt. In the face of more complicated relationships that admit multiplicative variation across dimensions, these additive effects are unsatisfactory to control for unobserved heterogeneity. This paper develops tools to control for unobserved heterogeneity in the form of interactive fixed-effects in linear models of multidimensional panel data. The methods developed herein can also be applied in the standard two-dimensional setting for potentially improved convergence rates.

To fix ideas consider linear parameter estimation in the following interactive fixed-effects model with three dimensions,

$$Y_{ijt} = X_{ijt}'\beta + \sum_{\ell=1}^{L} \varphi_{i\ell}^{(1)} \varphi_{j\ell}^{(2)} \varphi_{t\ell}^{(3)} + \varepsilon_{ijt}, \tag{1}$$

where all terms in $\sum_{\ell=1}^{L} \varphi_{i\ell}^{(1)} \varphi_{j\ell}^{(2)} \varphi_{t\ell}^{(3)}$ are unobserved and L is bounded and fixed. The object L is complicated and often difficult to calculate. However, for purposes of this analysis it is more important to know what is called the multilinear rank, defined in Section 2.1, which is taken as known here. Heuristics to estimate the multilinear rank are discussed, however in the context of covariates, estimation of interactive fixed-effect rank is still an active area of research. Simulations suggest estimation with more than L components performs well. Reducing the problem to three dimensions is without loss of generality for the methods considered herein. Additive fixed-effects are omitted for brevity but are subsumed by the interactive fixed-effect term or can be removed with a simple within transformation. Let X_{ijt} be arbitrarily correlated with the unobserved interactive fixed-effects term, $\sum_{\ell=1}^{L} \varphi_{i\ell}^{(1)} \varphi_{j\ell}^{(2)} \varphi_{t\ell}^{(3)}$, but uncorrelated with the noise term, ε_{ijt}. The challenge to estimating β is isolating variation in X_{ijt} that is not correlated with the interactive fixed-effects term. This paper develops the multidimensional group fixed-

[1] A non-exhaustive list of related examples can be found in the introduction of Matyas (2017) in trade, housing and prices, migration, country productivity and consumer price setting.

effects and kernel weighted transformations to project out this unobserved heterogeneity and also shows settings were standard factor methods work well. The kernel weighted within transformation is a novel contribution to the best of the author's knowledge. The group fixed-effects method uses similar clustering techniques from Bonhomme, Lamadon and Manresa (2022) and the within-cluster transformation in Freeman and Weidner (2022).

This paper makes two main contributions to the literature. The first is to show that the three or higher dimensional model can be couched in a standard two-dimensional panel data model and to derive sufficient conditions for consistency using these methods. The second contribution is to introduce kernel weighted fixed-effects methods and extend group fixed-effects methods to the multidimensional setting. The asymptotic results show that under certain conditions the group and kernel weighted fixed-effects can achieve the parametric rate of convergence, which are yet to be proven in the three dimensional case with current methods. The simulation results corroborate these theoretical findings and an empirical application that estimates the demand elasticity of beer demonstrates how these methods work in practice.

The novel estimators proposed in this paper can be described as simple extensions to the usual within transformation. This makes the estimator palatable for practitioners, and easy to implement. Consider additive fixed-effects of the form $a_{ij} + b_{it} + c_{jt}$. This can be projected using the transformation,

$$\dot{Y}_{ijt} = Y_{ijt} - \bar{Y}_{\cdot jt} - \bar{Y}_{i\cdot t} - \bar{Y}_{ij\cdot} + \bar{Y}_{\cdot\cdot t} + \bar{Y}_{\cdot j\cdot} + \bar{Y}_{i\cdot\cdot} - \bar{Y}_{\cdot\cdot\cdot}, \tag{2}$$

applied equivalently to X_{ijt}, where the bar variables denote the average taken over the "dotted" index for the entire sample. That is, $\bar{Y}_{\cdot jt} := \frac{1}{N_1}\sum_{i=1}^{N_1} Y_{ijt}$, $\bar{Y}_{\cdot\cdot t} := \frac{1}{N_1 N_2}\sum_{i=1}^{N_1}\sum_{j=1}^{N_2} Y_{ijt}$, etc. where emphasis is on the fact the means are taken uniformly over observations. In the presence of interactive fixed-effects, it turns out the simple within transformation demeans each fixed-effect term to leave,

$$\sum_{\ell=1}^{L}\left(\varphi_{i\ell}^{(1)} - \bar{\varphi}_{\ell}^{(1)}\right)\left(\varphi_{j\ell}^{(2)} - \bar{\varphi}_{\ell}^{(2)}\right)\left(\varphi_{t\ell}^{(3)} - \bar{\varphi}_{\ell}^{(3)}\right)$$

which is clearly not controlled for if the φ terms admit heterogeneity across observations. Of course, the simple within transformation was not designed to control for interactions, but this serves as a useful heuristic.

Consider instead a simple extension of the within transformation that uses weighted means instead of uniform means. With an abuse of notation, consider,

$$\widetilde{Y}_{ijt} = Y_{ijt} - \bar{Y}_{i^*jt} - \bar{Y}_{ij^*t} - \bar{Y}_{ijt^*} + \bar{Y}_{i^*j^*t} + \bar{Y}_{i^*jt^*} + \bar{Y}_{ij^*t^*} - \bar{Y}_{i^*j^*t^*} \tag{3}$$

where the bar variables combined with the star indices denote weighted means for that observation. For example, $\bar{Y}_{i^*jt} = \sum_{i'} w_{i,i'} Y_{i'jt}$ for weights across i' for each i, that need not be symmetric. This projects out the more generic fixed-effect, $\sum_{t'} w_{t,t'} a_{ijt'} + \sum_{j'} w_{j,j'} b_{ij't} + \sum_{i'} w_{i,i'} c_{i'jt}$. In turn, the remainder from the interactive fixed-effects term is,

$$\sum_{\ell=1}^{L} \left(\varphi_{i\ell}^{(1)} - \sum_{i'} w_{i,i'} \varphi_{i'\ell}^{(1)}\right) \left(\varphi_{j\ell}^{(2)} - \sum_{i'} w_{j,j'} \varphi_{j'\ell}^{(2)}\right) \left(\varphi_{t\ell}^{(3)} - \sum_{i'} w_{t,t'} \varphi_{t'\ell}^{(3)}\right).$$

Under regularity conditions, when each φ term admits heterogeneity, the weighted within transformation can project out the interactive fixed-effects if appropriate weights are used. The same can be true if the weighted means are replaced with cluster means for appropriate cluster assignments, similar to a group fixed-effects estimator. Hence, with a relatively small change to how the within transformation is performed, much more general fixed-effects can be considered in the linear model.

The model for interactive fixed-effects has precedent in the standard two-dimensional panel data setting. For instance take the model considered in Bai (2009) and similar to Pesaran (2006),

$$Y_{it} = X_{it}'\beta + \sum_{\ell=1}^{L} \lambda_{i\ell} f_{t\ell} + e_{it}. \tag{4}$$

In that setting, Bai (2009) show that the interactive term $\sum_{\ell=1}^{L} \lambda_{i\ell} f_{t\ell}$ also sufficiently captures variation in additive individual and time effects without the need to specify these separately, so these are again naturally omitted. For multidimensional applications the problem in (1) can be simply transformed to a two dimensional problem and estimated as (4) directly using the transformed data. Indeed, Section 3.1 displays useful preliminary convergence rates for this approach. However, convergence rates can suffer severely from the over-parametisation implied by (4). Without strong assumptions on the sparsity of the fixed-effects, only a very slow rate of convergence can be shown for this approach. However, when this approach is used to construct preliminary estimators of the fixed-effects, the usual parametric rate of consistency is possible when these are combined with the novel grouped or weighted within transformations. Kuersteiner and Prucha (2020) consider a similar dimension specific projection across the time dimension in the presence of spatial and serial correlation, in particular the Helmert transformation used in Supplement D.5.2.

On top of this slow convergence rate for the two dimensional transformation, finite sample bias may also arise when L is large and only a subset of the unobserved heterogeneity parameters are low-dimensional. For small bias in the estimates of β, transforming the multidimensional array to a matrix then estimating (4) requires either: (a) all the multidimensional fixed-effects are low-dimensional, or; (b) that a subset of the multidimensional fixed-effects are low-dimensional and the analyst knows which ones are. The requirement that the analyst has this knowledge can be highly restrictive. Alternatively, whilst the within-cluster and kernel weighted transformations analysed in this paper also require that a subset of the fixed-effect parameters are low-dimensional, the analyst does not need to know which ones are. In this sense, the novel within-cluster and kernel weighted estimators are robust. A concrete example is considered in a simulation exercise.

The demand elasticity for beer application uses Dominick's supermarket data from the Chicago area from 1991-1995, where price and quantity vary over product, store, and fortnight. For comparison, instrumental variables analysis is used to estimate elasticities, with the price of barley used as an instrument. IV estimates show demand is strongly negatively elastic (-2.15), however estimates are very imprecise because the instrument only varies over one dimension, time. Estimates from the weighted within transformation show similarly that demand is downward sloping and elastic (-2.83), but with much better precision. Estimates from existing two-dimensional methods are very sensitive to how the data is transformed into a two-dimensional dataset. The estimates from the weighted-within transformation are similar but more moderate than the own-price elasticities in Table 1 from Hausman, Leonard and Zona (1994).

The technical component of this paper is highly related to the numerical analysis literature on low-rank approximations of multidimensional arrays. As pointed out in De Silva and Lim (2008), the optimisation problem of finding low-rank approximations in the tensor setting is not well-posed, hence most results in this literature rely on numerical evidence. See Kolda and Bader (2009) for a summary of the multidimensional array decomposition problem and Vannieuwenhoven, Vandebril and Meerbergen (2012); Rabanser, Shchur and Günnemann (2017) for examples of numerical results. As such, it is necessary to innovate on this tensor low-rank problem to find appropriate analytical results. To this end, this paper utilises well-posed components of two-dimensional sub-problems for use in nuisance parameter applications. This application has the advantage that they only require the multidimensional array of fixed-effects to be projected out at a sufficient asymptotic rate, and do not attempt to directly solve the low-rank tensor problem. It is worth a note that Elden and Savas (2011), along with related papers, suggest a reformulation of the low multilinear rank problem that may have promising applications in econometrics.

Under sufficient regularity conditions, the methods considered in this paper may also control for variation from arbitrary smooth functions of the low dimensional fixed-effects. Similar to that considered in Zeleneev (2020) and Freeman and Weidner (2022), the functional representation of model (1) could be,

$$Y_{ijt} = X_{ijt}'\beta + h(\varphi_i^{(1)}, \varphi_j^{(2)}, \varphi_t^{(3)}) + \varepsilon_{ijt},$$

for vector-valued $\varphi_i^{(1)}$, $\varphi_j^{(2)}$ and $\varphi_t^{(3)}$. The set of fixed-effects to be transformed by the function $h(\cdot, \cdot, \cdot)$ could also extend to fixed-effects over multiple indices, e.g. α_{ij} from above. It should be noted that the setting considered in Zeleneev (2020) requires that the transformation is non-smooth, and it is not trivial to see that a "within-type" transformation will sufficiently project this type of heterogeneity. Formal considerations of this general model are left for future work. Convergence rates for β estimation are also likely to be slower in the functional setting.

The methods proposed herein may also serve to alleviate convergence issue in certain two-dimensional panel data models when N and T are not proportional. Most of the interactive fixed effects literature requires that the ratio $N/T \to \rho \in (0, \infty)$, that is, that N, T grow at the same rate. As a workaround, consider splitting the larger of the N or T dimension into sub-dimensions. For example, if $T = o(N)$, such that $N >> T$, then split N into as many sub-dimensions as necessary. If $N = O(T^\alpha)$ for $\alpha \in (1, M)$, then split N into roughly α equal parts. For say $\alpha = 2$, split the first dimension into indices i, j to leave the interactive fixed-effect term,

$$\sum_{\ell=1}^{L} \lambda_{ij\ell} f_{t\ell} = \sum_{\ell=1}^{L} \left(\sum_{\ell'=1}^{L'} \lambda^{(1)}_{i\ell'\ell} \lambda^{(2)}_{j\ell'\ell} \right) f_{t\ell},$$

where $\lambda^{(1)}_{i\ell'\ell}$ and $\lambda^{(2)}_{j\ell'\ell}$ simply denote the superimposed fixed-effect parameters for the new sub-dimensions. If there is sufficient sparsity in the first dimension fixed-effect parameter, $\lambda_{ij\ell}$, then L' can be bounded and fixed, and the interactive fixed-effect term is synonymous with the multidimensional interactive fixed-effects in (1). This is of course an additional sparsity assumption on the fixed-effects, but may serve as a useful practical solution to settings with vastly different N and T.

Menzel (2021) consider a special case of multidimensional data for bootstrapping methods where the data is D-adic. That is, each dimension of the data refers to the same set of observations, like a network graph where each index refers to an individual in the network. An example of the multidimensional version of this could be a binary indicator of a three step path, $Y_{ijk} = G_{ij} G_{jk}$, detailing if there exists a path from i to k. In any case, the type of multidimensional data considered in that work is a distinct special case of the type of data structures considered in this paper.

The paper is organised as follows. Section 2 introduces the model, and notation and preliminaries; Section 3 details the estimators and associated assumptions with convergence results; Section 4 discusses the convergence results along with some alternative assumptions, and further motivates the estimation approach; Section 5 displays the simulation results; Section 6 shows the beer demand estimation empirical application; and Section 7 concludes.

2 Model

Let β^0 denote the true parameter value for the slope coefficients. The model in full dimensional generality is, [2]

$$\boldsymbol{Y} = \sum_{k=1}^{K} \boldsymbol{X}_k \beta_k^0 + \boldsymbol{\mathcal{A}} + \boldsymbol{\varepsilon}, \tag{5}$$

where $\boldsymbol{Y}, \boldsymbol{X}_k, \boldsymbol{\varepsilon} \in \mathbb{R}^{N_1 \times N_2 \times \cdots \times N_d}$. $\boldsymbol{\mathcal{A}} = \sum_{\ell=1}^{L} \varphi_\ell^{(1)} \circ \cdots \circ \varphi_\ell^{(d)}$ where $\varphi_\ell^{(n)} \in \mathbb{R}^{N_n}$ for each $n = 1, \ldots, d$ and "$\circ$" is the outer product. L is bounded and fixed. $\boldsymbol{\varepsilon}$ is a noise term uncorrelated with all $\boldsymbol{X}_k$ and all unobserved fixed-effects terms. Take $i_n \in \{1, \ldots, N_n\}$ for all $n \in \{1, \ldots, d\}$ as the dimension specific index, where N_n is the sample size of dimension n. The regressors $\boldsymbol{X}_k$ may be arbitrarily correlated with $\boldsymbol{\mathcal{A}}$. Throughout this paper all dimensions are considered to grow asymptotically, that is $N_n \to \infty$ for all n, however, for the asymptotic theory only a subset of two dimensions need to grow asymptotically.

Model (5) can be seen as a natural extension of the Bai (2009) model to three (or more) dimensions with the $\boldsymbol{\mathcal{A}}$ term interpreted as a "higher-dimensional" factor stucture. Similar to this strain of the literature, all terms in $\boldsymbol{\mathcal{A}}$ are considered fixed nuisance parameters. There are potentially many extensions to the factor model setting in Bai (2009) to the higher dimension case. This paper starts with what seems the most natural extension.

The term $\mathcal{A}$ may also incorporate additive fixed effects that vary in any strict subset of the dimensions. For example, in the three dimensional setting one may want to control for the additive terms, $a_{ij}+b_{it}+c_{jt}$. These can be controlled for using $L = \min\{N_1, N_2\}+\min\{N_1, N_3\}+\min\{N_2, N_3\}$, with the first $\min\{N_1, N_2\}$ terms $\sum_{\ell=1}^{\min\{N_1,N_2\}} \varphi_{i\ell}^{(1)} \varphi_{j\ell}^{(2)} = a_{ij}$ by setting $\varphi_{t\ell}^{(3)} = 1$ for $\ell = 1, \ldots, \min\{N_1, N_2\}$, and so on for the b_{it} and c_{jt}. These could also be controlled for directly using the standard within-transformation before considering the model in (5).

This paper comprises of two main modelling approaches. The first is to embed the multidimensional model into a standard panel data model by simply flattening all arrays into matrices. The second approach uses weighted differences across each dimension to reduce each $\varphi^{(n)}$ component of $\mathcal{A}$ separately for each n. For this reason the model assumptions are split out and stated in Section 3 alongside each estimation approach.

[2] For example, in index notation this model can be written as,

$$Y_{i_1,i_2,\ldots,i_d} = \sum_{k=1}^{K} X_{i_1,i_2,\ldots,i_d;k} \beta_k^0 + \mathcal{A}_{i_1,i_2,\ldots,i_d} + \varepsilon_{i_1,i_2,\ldots,i_d}$$

with $\mathcal{A}_{i_1,i_2,\ldots,i_d} = \sum_{\ell=1}^{L} \varphi_{i_1\ell}^{(1)} \cdots \varphi_{i_d\ell}^{(d)}$.

2.1 Notation and preliminaries

For a d-order tensor, $\mathbf{A}$, a factor-n flattening, denoted as $\mathbf{A}_{(n)}$, is the rearrangement of the tensor into a matrix with dimension n varying along the rows and the remaining dimensions simultaneously varying over the columns. That is, $\mathbf{A}_{(n)} \in \mathbb{R}^{N_n \times N_{n+1} N_{n+2} \ldots N_1 \ldots N_{n-1}}$. The Frobenius norm, $\| \cdot \|_F$, of a matrix or tensor is the entry-wise norm, $\|\mathbf{A}\|_F^2 = \sum_{i_1=1}^{N_1} \cdots \sum_{i_d=1}^{N_d} A_{i_1 \ldots i_d}^2$. The spectral norm, denoted $\| \cdot \|$, is the largest singular value of a matrix. For a d-order tensor, $\mathbf{A}$, the multilinear rank, denoted $\mathbf{r}$, is a vector of matrix ranks after factor-n flattening in each dimension, with each component of this vector $r_n = rank\big(\mathbf{A}_{(n)}\big)$. Tensor rank, different to multilinear rank, is defined as the least number of outer products of vectors to replicate the tensor. That is, for tensor $\mathbf{A}$ and vectors $u_\ell^{(n)} \in \mathbb{R}^{N_n}$, tensor rank is the smallest L such that $\mathbf{A} = \sum_{\ell=1}^{L} u_\ell^{(1)} \circ \cdots \circ u_\ell^{(d)}$, where $\circ$ is the outer product of a vector. The notation $a \lesssim b$ means the asymptotic order of a is bounded by the asymptotic order of b.

The n-mode product between a tensor $\mathbf{A}$ and matrix B is denoted $\mathbf{A} \times_n B$ and has elements

$$(\mathbf{A} \times_n B)_{i_1,\ldots,j,\ldots,i_d} = \sum_{i_n=1}^{N_n} A_{i_1,\ldots,i_n,\ldots,i_d} B_{j,i_n},$$

which is equivalent to saying the flattening $(\mathbf{A} \times_n B)_{(n)} = B\mathbf{A}_{(n)}$. This can be referred to as "hitting" the tensor $\mathbf{A}$ with matrix B in the n^{th} dimension, though this terminology is only stated to help with understanding of the definition.

The singular value decomposition is used in both estimation approaches in this paper so it is important to understand some of its properties. The singular value decomposition of a matrix, $A \in \mathbb{R}^{N_1 \times N_2}$ is

$$A = U\Sigma V' = \sum_{r=1}^{\min\{N_1,N_2\}} \sigma_r u_r v_r', \tag{6}$$

where U is the matrix of left singular vectors, u_r, V is the matrix of right singular vectors, v_r, and Σ is a diagonal matrix of singular values, σ_r, with values running in descending order down the diagonal. For a rank-r matrix, the first r entries on the diagonal of Σ are strictly positive and the remaining entries are zero.

Take the approximation problem,

$$\min_{A'} \|A - A'\|_F \text{ such that } rank(A') = k. \tag{7}$$

It is well known from the Eckart-Young-Mirsky theorem that the solution to this approximation problem is the first k terms of the singular value decomposition, i.e. $\sum_{r=1}^{k} \sigma_r u_r v_r'$. The Eckart-Young-Mirsky theorem effectively picks out the row and column subspaces that best explain variation in the matrix A as the leading columns of the matrix U, respectively of V. The sum of squared error at the minimiser is thus $\sum_{r=k+1}^{\min\{N_1,N_2\}} \sigma_r^2$. This is commonly called a low-rank approximation and forms the cornerstone for estimation of unobserved heterogeneity in the factor model and interactive fixed-effects models in Bai and Ng (2002); Bai (2009); Moon and Weidner (2015) amongst others.

The Eckart-Young-Mirsky theorem, however, does not extend to the three or higher dimensional setting, see De Silva and Lim (2008) for details. To avoid this complication, the multidimensional problem can be translated to the two-dimensional setting to utilise the Eckart-Young-Mirsky theorem, or the fixed-effects parameters need to be shrunk separately, as is done with the group fixed-effects and kernel methods.

3 Estimation

This section details the three estimation approaches used. The first subsection details how to apply standard two dimensional estimators to the problem and the assumptions required for consistent estimation. The second and third subsections detail the group fixed-effect and kernel weighted approaches and the required assumptions for consistency in those settings.

3.1 Matrix low-rank approximation estimator

This section provides a description of some matrix methods that can be applied directly to the multidimensional model and stipulates the assumptions required for consistency. Note that Kapetanios, Serlenga and Shin (2021) employ a similar approach for three-dimensional arrays in conjunction with the Pesaran (2006) common correlated effects estimator. Babii, Ghysels and Pan (2022) employ a similar matricisation procedure as that detailed below, but are interested in inference on the fixed-effect parameters.

Consider recasting the multidimensional array problem into a two dimensional panel problem by flattening Y and X in the n-th dimension,

$$Y_{(n)} = X'_{(n)}\beta^0 + \varphi^{(n)}\Gamma'_n + \varepsilon_{(n)}$$

where $Y_{(n)}, X_{(n)}, \varepsilon_{(n)} \in \mathbb{R}^{N_n \times \prod_{n' \neq n} N_{n'}}$, $\varphi^{(n)}$ is an $N_n \times r_n$ matrix and Γ_n is an $\prod_{n' \neq n} N_{n'} \times r_n$ matrix that accounts for variation in the remaining $\varphi^{(n')}$ for all $n' \neq n$. The term r_n is indexed by the dimension n because it may vary non-trivially according to the flattened dimension. It should then be apparent that this is exactly the model described in (4), that is, the standard linear model with factor structure unobserved heterogeneity as studied in Bai (2009).

The two-dimensional estimator for a given flattening, n, optimises the following objective function,

$$R(\beta, \widehat{r}_n, n) = \min_{\substack{\varphi^{(n)} \in \mathbb{R}^{N_n \times \widehat{r}_n}, \\ \Gamma_n \in \mathbb{R}^{\prod_{n' \neq n} N_{n'} \times \widehat{r}_n}}} \left\| Y_{(n)} - X'_{(n)}\beta - \varphi^{(n)}\Gamma'_n \right\|_F^2 . \tag{8}$$

Then $\widehat{\beta}^{2D}_{(n)} = \text{argmin}_\beta R(\beta, \widehat{r}_n, n)$ is the slope estimate for the two-dimensional setup. The analyst must choose both the dimension to flatten in, n, and the rank of the estimated interactive fixed-effects term, $\widehat{r}_n$. It is well known that the minimum in (8) is achieved using the leading $\widehat{r}_n$ terms from the singular value decomposition of the error term, $Y_{(n)} - X'_{(n)}\beta$. This gives $\widehat{\varphi}^{(n)}$ as the first $\widehat{r}_n$ columns of $\widehat{U}\widehat{\Sigma}$ and $\widehat{\Gamma}_n$ as the first $\widehat{r}_n$ columns of $\widehat{V}$ where $\widehat{U}$, $\widehat{\Sigma}$ and $\widehat{V}$ are the terms from (6) of the singular value decomposition of $Y_{(n)} - X'_{(n)}\beta$. Because this error term is a function of β, an iteration is naturally required between estimating β and finding the singular value decomposition of the error term. This is a well studied iteration procedure, for convergence details see Bai (2009); Moon and Weidner (2015).

In the following assumptions let $\widehat{r}_n$ be the estimated number of factors for the (n)-flattening of the regression line when applying the least square methods in (8). Also, let $\mathcal{L} \subset \{1, \ldots, d\}$ be a non-empty subset of the dimensions. In the following, the multilinear rank of $\boldsymbol{\mathcal{A}}$ is restricted such that it is low-rank along at least one of the flattenings.

Assumption 1 (Bounded norms of covariates and exogenous error)**.**

(i). $\left\| X_k \right\|_F = O_p\left(\prod_{n=1}^d \sqrt{N_n} \right)$ *for each* k

(ii). $\left\| \varepsilon_{(n^*)} \right\| = O_p\left(\max\{\sqrt{N_{n^*}}, \prod_{m \neq n^*} \sqrt{N_m}\} \right)$ *for each* $n^* \in \mathcal{L}$

Assumption 2 (Weak exogeneity)**.** $\text{vec}(X_k)'\text{vec}(\varepsilon) = O_p\left(\prod_{n=1}^d \sqrt{N_n} \right)$ *for each* k

Assumption 3 (Low multilinear rank)**.** *For some positive integer,* c*,* $r_{n^*} < c$ *for all* $n^* \in \mathcal{L}$*, where* r_n *is the* n^{th} *component of the multilinear rank of* $\boldsymbol{\mathcal{A}}$*.*

Assumption 4 (Non-singularity)**.** *Let* $\sigma_s(A)$ *be the* s^{th} *singular value for a matrix* A*. Consider linear combinations* $\delta_{n^*} \cdot X_{(n^*)} = \sum_k \delta_{n^*,k} X_{(n^*),k}$*. For each dimension* $n^* \in \mathcal{L}$ *that satisfies Assumption 3, then for* $K \times 1$ *unit vector* δ_{n^*}*,*

$$\min_{\{\delta_{n^*} \in \mathbb{R}^K, \|\delta_{n^*}\|=1\}} \sum_{s = r_{n^*} + \widehat{r}_{n^*} + 1}^{\min\{N_{n^*}, \prod_{m \neq n^*} N_m\}} \sigma_s^2 \left(\frac{(\delta_{n^*} \cdot X_{(n^*)})}{\prod_n \sqrt{N_n}} \right) > b > 0 \qquad wpa1.$$

Assumptions 1, 2 and 4 are standard regularity assumptions already well established in the literature, e.g. see Moon and Weidner (2015). Assumption 1.(i) ensures that the covariates have bounded norms, for example having bounded second moments. Assumption 1.(ii) allows for some weak correlation across dimensions, see Moon and Weidner (2015), or is otherwise implied if the noise terms are independently distributed with bounded fourth moments, see Latała (2005). Assumption 2 is implied if $X_{i_1,i_2,\ldots,i_d;k}\varepsilon_{i_1,i_2,\ldots,i_d}$ are zero mean, bounded second moment and only admits weak correlation across dimensions for each $k = 1, \ldots, K$. Assumption 4 simply states that, after factor projection, the set of covariates still collectively admit full-rank variation.

Assumption 3 is new and asserts that there exists at least one flattening of the interactive term, $\boldsymbol{\mathcal{A}}$, that is low-dimensional or simply low-rank. This requires that at least one of the unobserved terms $\varphi^{(n)}$ is low dimensional. Note that not all dimensions must satisfy Assumption 3 for the below result. If the correct dimension is chosen then variation from the interactive term can be sufficiently projected out using the factor model approach. This makes up the statement of the following Proposition.

Proposition 1. *Let $\widehat{\beta}^{2D}_{(n^*)}$ be the estimator from Bai (2009) after first flattening along dimension $n^* \in \mathcal{L}$. If Assumptions 1-4 hold, the subset $\mathcal{L}$ is non-empty, and the estimated number of factors $\widehat{r}_{n^*} \geq r_{n^*}$, then, for each $n^* \in \mathcal{L}$ satisfying Assumption 3,*

$$\left\| \widehat{\beta}^{2D}_{(n^*)} - \beta^0 \right\| = O_p\left(\frac{1}{\sqrt{\min\{N_{n^*}, \prod_{n \neq n^*} N_n\}}} \right). \tag{9}$$

Proposition 1 follows directly from Moon and Weidner (2015) since the flattening procedure reduces the problem to the standard linear interactive fixed-effects model. Notice that this result only applies to estimates in the dimension(s) that satisfy the low-rank assumption in Assumption 3. That is, implicit in Proposition 1 is that the analyst has chosen the correct dimension to flatten over when reformulating the problem as a two-dimensional panel. The constraint $\widehat{r}_{n^*} \geq r_{n^*}$ can also be changed to $\widehat{r}_{n^*} \geq c$, however, this is more conservative than required for the statement of the result. This constraint does not require knowledge of r_{n^*}, just that the number of estimated factors is greater than or equal the true number.

The estimation procedure from Proposition 1 can also be augmented to flatten over multiple indices. For instance, the analyst may flatten such that both the rows and columns in the matrix contain multiple indices from the original array. Of course, this augmentation makes Assumption 3 harder to satisfy as it requires multiple parameters to vary in low-dimensional space. To see this take the tensor $\boldsymbol{\mathcal{A}}$ flattened over the first two indices as $\mathcal{A}_{(1,2)} \in \mathbb{R}^{N_1 N_2 \times \prod_{n \notin \{1,2\}} N_n}$. If the parameters $\varphi^{(n)}$ for $n = 3, \ldots, d$ are high-dimensional, Assumption 3 is only satisfied when both $\varphi^{(1)}$ and $\varphi^{(2)}$ and their product space is low-dimensional. Clearly this is more restrictive than requiring only one of the parameter spaces to be low-dimensional. However, flattening along multiple dimensions can improve the convergence rate in Proposition 1 to $O_p\left(\frac{1}{\sqrt{\min\{N_1 N_2, \prod_{n \notin \{1,2\}} N_n\}}}\right)$, so there are benefits if this more restrictive assumption can be made. Further discussion of the matrix method results are relegated to Section 4.1, in particular some avenues to choosing the dimension to flatten over.

3.2 Kernel weighted fixed-effects

Let $\widehat{\varphi}_{i_n}^{(n)}$ generically denote a proxy measure for unit i_n in dimension n that may be known or estimated. The use of this notation will become clear in the statement of Proposition 2 and in discussion of how to estimate these proxy measures in Section 4.2. Let $\mathcal{W}$ be an ordered set of weight matrices, where the n^{th} item $W_n \in \mathbb{R}^{N_n \times N_n}$ has elements,

$$W_{n,i_n j_n} := \frac{k\left(\frac{1}{h_n}\left\|\widehat{\varphi}_{i_n}^{(n)} - \widehat{\varphi}_{j_n}^{(n)}\right\|\right)}{\sum_{i'_n=1}^{N_n} k\left(\frac{1}{h_n}\left\|\widehat{\varphi}_{i_n}^{(n)} - \widehat{\varphi}_{i'_n}^{(n)}\right\|\right)}, \tag{10}$$

where k is a kernel function, and h_n is a bandwidth parameter. The weighted-within transformation in (3) can be generalised with the following series of n-mode products, $\mathbf{Y} \times_1 M_1 \times_2 M_2 \times_3 \cdots \times_d M_d$, likewise also on each $\mathbf{X}_k$, where $M_n = \mathbb{I}_{N_n} - W_n$ and $\times_n$ is the n-mode product defined in Section 2.1. This projection may not optimise any specific objective function - it is designed to sufficiently project out variation in the fixed-effect term. In this sense, the following theoretical analysis is viewed in the light of an asymptotic bias problem. Let $\widehat{\beta}_{KER,\mathcal{W}}$ be the pooled OLS estimator after using weights W_n in the weighed-within transformation on each observed variable. The technical assumptions and results in this section relate to the estimator $\widehat{\beta}_{KER,\mathcal{W}}$.

If proxy measures are estimated from the error term $\mathbf{Y} - \sum_{k=1}^{K} \mathbf{X}_k \beta_k$, then there is an iteration between slope estimation and estimation of kernel weights, much like in the matrix method presented already. This can also be optimised over a grid space for β if β is low-dimensional. Alternatively, estimates used to calculate weights may come from the matrix method procedure, and the weighed-within estimator used as a debias.

Assumption 5 (Kernels)**.** *The kernel function, $k(\cdot)$ is, (i). Non-negative, non-increasing, and integrable (ii). $\int k(u)du = 1$ (iii). $k(-u) = k(u)$ (iv). For $a > 0$ and $h > 0$, $k(a/h)a \lesssim O(h)$ as $h \to 0$.*

Assumption 5.(i)-(iii) are standard kernel function restrictions. Assumption 5.(iv) refers to a bandwidth parameter, h, and restricts the kernels to penalise distance at a rate equal to or faster than $O(h/a)$. For consistency using the kernel methods, the sequence $h \to 0$ is considered.

Assumption 6 (Regularity of proxy measures)**.** *Let $\widehat{\varphi}_{i_n}^{(n)} \in \widehat{\Phi}_n$ and define $M_n^h\left(\widehat{\varphi}_{i_n}^{(n)}\right)$ as*

$$M_n^h\left(\widehat{\varphi}_{i_n}^{(n)}\right) := \sum_{j_n=1}^{N_n} \mathbb{1}\left(\widehat{\varphi}_{j_n}^{(n)} \in B_{he}\left(\widehat{\varphi}_{i_n}^{(n)}\right)\right),$$

where $B_{he}(x)$ is an he-neighbourhood around x. Then for each dimension n, for any $e > 0$, as

$h \to 0$ such that $N_n h \to \infty$,

$$\operatorname*{plim}_{N_n \to \infty} \frac{M_n^h\left(\widehat{\varphi}_{i_n}^{(n^*)}\right)}{N_n h} = c_{i_n}^{(n)} \in (0, h^{-1}]$$

for all $\widehat{\varphi}_{i_n}^{(n)} \in \widehat{\Phi}_n$. Further, let $\widehat{\varphi}_{i_n}^{(n)} \in \widehat{\Phi}_n$ have a bounded second moment.

Assumption 6 is a restriction on how the probability space for the fixed-effects proxy parameter space is sampled from. The assumption is stated for vector-valued $\widehat{\varphi}_{i_n}^{(n)} \in \widehat{\Phi}_n$. However, the weighted-within transformation can be implemented iteratively, so the assumption can be relaxed to an element-wise restriction on each component of $\widehat{\varphi}_{i_n}^{(n)} \in \widehat{\Phi}_n$, but leads to a significantly more complicated theory. See Appendix A.2 for a treatment of the iterative estimator that deals with this less restrictive assumption. Assumption A.1 and Lemma A.3 in the Appendix also deal with relaxing Assumption 6 to a mean bound, rather than the uniform bound as currently stated. In this sense, Assumption 6 is viewed as a simplifying assumption. Chapter 5 from Györfi, Kohler, Krzyzak, Walk et al. (2002) discusses many similar kernel regression results with lower level assumptions that imply similar conditions to Assumption 6. The low level restriction in Assumption 6 on the proxies can be used to show a higher-level condition on the sampling of the kernel function, see Lemma A.1 in the Appendix. The condition in Assumption 6 ensures that, asymptotically, neighbourhoods around each proxy measure are non-empty, with at least order $N_n h$ number of observations in the neighbourhood. This is to make sure the estimator has well-defined weights, that do not degenerate for each i_n. This restriction ensure that mean square error results can be applied to the proxy measure space in the case when they are estimated.

Assumption 7 (Regularity conditions). *Let $\check{T}_{i_1,\dots,i_d}$ be the entries of tensor $\mathbf{T}$ after the kernel weighted fixed-effects from the minimiser of* (11) *are differenced out. Then,*

(i). $\left(\frac{1}{\prod_n N_n} \sum_{i_1} \cdots \sum_{i_d} \check{X}_{i_1,\dots,i_d} \check{X}'_{i_1,\dots,i_d}\right)$ *converges in probability to a nonrandom positive definite matrix as $N_1, \dots, N_d \to \infty$.*

(ii). $\frac{1}{\prod_n N_n} \sum_{i_1} \cdots \sum_{i_d} \check{X}_{i_1,\dots,i_d} \varepsilon_{i_1,\dots,i_d} = O_p\left(\frac{1}{\sqrt{\prod_n N_n}}\right)$.

Assumption 7.(i) is very similar to Assumption 4 except that here full rank is required after the weighted-within projection rather than the factor projection. Assumption 7.(ii) is an exogeneity condition that requires weak exogeneity in the covariates after the weighted-within transformation, which can be viewed as similar to Assumption 2. This is stricter than Assumption 2 because the noise term ε can foreseeably impact weights if weights are estimated from functionals of a residual term. This limitation is alleviated by, for instance, making sure weights are based on variables extraneous to the regression line, hence independent of ε. Likewise, independence can be achieved with a sample split, such as that proposed in Freeman and Weidner (2022). Here weights could be formed from estimates made in one part of the split, and differenced out with the weighted-within transformation from the other part of the split.

Proposition 2 (Upper bound on kernel weighted estimator)**.** *Let Assumptions 5-7 hold. Let* $\frac{1}{N_{n^*}} \sum_{i_{n^*}} \left\| \varphi^{(n^*)}_{i_{n^*}} - \widehat{\varphi}^{(n^*)}_{i_{n^*}} \right\|^2 = O_p(C_{n^*}^{-2})$ *for* $n^* \in \mathcal{M}'$ *with* $C_{n^*}^{-2} \to 0$*, and* $\frac{1}{N_{n'}} \sum_{i_{n'}} \left\| \varphi^{(n')}_{i_{n'}} - \widehat{\varphi}^{(n')}_{i_{n'}} \right\|^2 = O_p(1)$ *for* $n' \notin \mathcal{M}'$*, where* $\mathcal{M}'$ *is a non-empty subset of dimensions. Let* h_n *be the bandwidth parameter satisfying Assumption 6. Then,*

$$\left\| \widehat{\beta}_{KER,\mathcal{W}} - \beta^0 \right\| = \sqrt{L} O_p \left(\prod_{n^* \in \mathcal{M}'} O_p\left(C_{n^*}^{-1}\right) + O_p\left(h_{n^*}\right) \right) + O_p \left(\prod_{n=1}^{d} \frac{1}{\sqrt{N_n}} \right).$$

For $h_n \lesssim O(C_n^{-1})$ *this reduces to*

$$\left\| \widehat{\beta}_{KER,\mathcal{W}} - \beta^0 \right\| = \sqrt{L} O_p \left(\prod_{n^* \in \mathcal{M}'} O_p\left(C_{n^*}^{-1}\right) \right) + O_p \left(\prod_{n=1}^{d} \frac{1}{\sqrt{N_n}} \right).$$

Proposition 2 shows that the convergence rate for the kernel estimator is bounded by the convergence rate of the proxy estimates. That is, as long as the bandwidth parameter approaches zero sufficiently fast, the kernel estimator converges at a rate no worse than the convergence of the proxies when proxies are estimations of the true parameter values at or slower than $\sqrt{N_n}$-convergence. This is expected and also a good result that the kernel method does not hinder the convergence rate from these proxies. These kernel methods do, however, suffer a curse of dimensionality since Assumption 6 becomes increasingly difficult to justify as the dimension of the fixed-effects increases. This can be addressed with an iterative estimator, iterating over the fixed-effects to be projected out, such that weights are only ever functions of one-dimensional objects. Discussions in Section 4.2 suggest $O_p(C_{n^*}^{-1})$ can be $O_p\big(1/\sqrt{N_{n^*}}\big)$ under regularity conditions imposed in Bai (2009). This shows the parametric rate is attainable if $\mathcal{M}' = \{1, \ldots, d\}$ and L is fixed and bounded.

Note that implicit in the statement of Proposition 2 is that the correct multilinear rank of the tensor of interactive fixed-effects is known, at least for the dimensions $\mathcal{M}'$. It is conjectured that this can be relaxed to be just that the upper bound on the multilinear rank is known, but this is left for further research. This would follow from the results in Moon and Weidner (2015), also used in Proposition 1.

3.2.1 Optimal kernel weighted fixed-effects

Consider as a diversion an estimator that does optimise an objective function, but adds significant complexity to the technical proofs. Let Δ be a d-list of $\bigtimes_{n=1}^{d} N_n$ tensors $\delta_n \in \mathbb{R}^{N_1 \times \cdots \times N_d}$.

For a given set of proxy measures and kernel function, the kernel weighted fixed-effects estimator optimises

$$S(\beta, \mathcal{W}) = \min_{\delta \in \Delta} \left\| \mathbf{Y} - \sum_{k=1}^{K} \mathbf{X}_k \beta_k - \sum_{n=1}^{d} \delta_n \times_n W_n' \right\|_F^2 . \tag{11}$$

Then, $\widehat{\beta}_{KEROpt,\mathcal{W}} := \text{argmin}_{\beta \in \mathbb{R}^K} S(\beta, \mathcal{W})$, where $KEROpt$ stands for kernel optimum. The notation $\times_n$ is the n-mode product defined in Section 2.1. Take M_n to be an $N_n \times N_n$ matrix defined as $\mathbb{I}_{N_n} - W_n(W_n'W_n)^\dagger W_n'$, where † is the Moore-Penrose generalised inverse. Then, the estimator can be formed by pooled OLS after the following series of n-mode products, $\mathbf{Y} \times_1 M_1 \times_2 M_2 \times_3 \cdots \times_d M_d$, likewise also on each $\mathbf{X}_k$. This sequentially differences out the weighted means from each dimension separately. Then the Frisch-Waugh-Lovell theorem straightforwardly applies.

Projection of these weighted fixed-effects is equivalent to using the weighted-within transformation in (3). To obtain the kernel optimum estimator, simply input weights $W_n(W_n'W_n)^\dagger W_n'$ to the weighted-within estimator. Simply using W_n as per the earlier $\widehat{\beta}_{KER,\mathcal{W}}$ estimator normalises weights to $[0, 1]$, which turns out to make technical proofs much simpler. It is expected, however, that the $\widehat{\beta}_{KEROpt,\mathcal{W}}$ estimator will have similar asymptotic properties.

3.3 Asymptotic Distribution

This section establishes an inference procedure for the kernel weighted fixed-effects model. The procedures here translate do not translate to the matrix low-rank approximation estimator, since convergence rates are too slow for that estimator. Proposition 2 establishes an upper bound on the kernel weighted fixed-effect estimator convergence rate that can be refined to exactly the parametric rate for the fixed-effect asymptotic bias component. To ensure the bias from the fixed-effect term converges sufficiently quickly, that is, faster than the parametric rate, a further correction is required. Below is an additional orthogonalisation that can be used to achieve this.[3]

Consider the conditional expectation, $\mathbb{E}\left(\boldsymbol{Y}|\boldsymbol{\mathcal{A}}\right) = \Gamma_Y$, $\mathbb{E}\left(\boldsymbol{X}|\boldsymbol{\mathcal{A}}\right) = \Gamma_X$, such that,

$$\boldsymbol{X} = \boldsymbol{\Gamma}_X + \boldsymbol{\eta}, \qquad \boldsymbol{\Gamma}_Y = \boldsymbol{\Gamma}_X \cdot \beta^0 + \boldsymbol{\mathcal{A}}, \qquad \boldsymbol{Y} - \boldsymbol{\Gamma}_Y = (\boldsymbol{X} - \boldsymbol{\Gamma}_X) \cdot \beta^0 + \boldsymbol{\varepsilon}, \tag{12}$$

with $\mathbb{E}\left(\boldsymbol{\eta}|\boldsymbol{\mathcal{A}}\right) = 0$, $\mathbb{E}\left(\boldsymbol{\varepsilon}|\boldsymbol{X}, \boldsymbol{\mathcal{A}}\right) = 0$. Note that the display for $\boldsymbol{X}$ is general, in that if $\boldsymbol{X}$ is unrelated to $\boldsymbol{\mathcal{A}}$, $\boldsymbol{\Gamma}_X = 0$. Hence, the common correlated effects display in (12) is only a representation, not an imposed model. Also note, if the correlation between $\boldsymbol{X}$ and $\boldsymbol{\mathcal{A}}$ is weak

[3] This method is a part of a joint paper with Dennis Kristensen. However, there is currently no working paper version to cite.

in the sense that the empirical mean of $\boldsymbol{\Gamma}_X$ converges to zero at arbitrary rate, then the following inference correction is not required and the convergence result from Proposition 2 is sufficient.

Then β^0 can be estimated as $\widehat{\beta}_{IC}$, where IC stands for inference correction, using the infeasible estimator,

$$\widehat{\beta}_{IC} = \left(\text{vec}_K(\boldsymbol{X} - \boldsymbol{\Gamma}_X)'\text{vec}_K(\boldsymbol{X} - \boldsymbol{\Gamma}_X)\right)^{-1} \left(\text{vec}_K(\boldsymbol{X} - \boldsymbol{\Gamma}_X)'\text{vec}(\boldsymbol{Y} - \boldsymbol{\Gamma}_Y)\right)$$

where $\text{vec}_K(\boldsymbol{X} - \boldsymbol{\Gamma}_X) \in \mathbb{R}^{\prod_n N_n \times K}$ is shorthand for the matrix of vectorised covariates, with each column the vectorised $(\boldsymbol{X}_k - \boldsymbol{\Gamma}_{X_k})$. The vec notation for $(\boldsymbol{Y} - \boldsymbol{\Gamma}_Y)$ is the standard vectorisation. This is so far infeasible because $\boldsymbol{\Gamma}_X$, and $\boldsymbol{\Gamma}_Y$ need to be estimated.

Consider estimates for $\boldsymbol{\Gamma}_X$, and $\boldsymbol{\Gamma}_Y$ as $\widehat{\boldsymbol{\Gamma}}_X$, and $\widehat{\boldsymbol{\Gamma}}_Y$, respectively, and $\widehat{\Omega}_X = \text{vec}_K(\boldsymbol{X} - \widehat{\boldsymbol{\Gamma}}_X)'\text{vec}_K(\boldsymbol{X} - \widehat{\boldsymbol{\Gamma}}_X)$. Then,

$$\begin{aligned}
\widehat{\Omega}_X^{-1}\text{vec}_K(\boldsymbol{X} - \widehat{\boldsymbol{\Gamma}}_X)'\text{vec}(\boldsymbol{Y} - \widehat{\boldsymbol{\Gamma}}_Y) &= \widehat{\Omega}_X^{-1}\text{vec}_K(\widehat{\boldsymbol{\eta}})'\text{vec}(\boldsymbol{X} \cdot \beta^0 + \boldsymbol{\mathcal{A}} + \boldsymbol{\varepsilon} - \widehat{\boldsymbol{\Gamma}}_X \cdot \widetilde{\beta} - \widehat{\boldsymbol{\mathcal{A}}}) \\
&= \widehat{\Omega}_X^{-1}\text{vec}_K(\widehat{\boldsymbol{\eta}})'\text{vec}_K(\widehat{\boldsymbol{\eta}}) \cdot \beta^0 \\
&\quad + \widehat{\Omega}_X^{-1}\text{vec}_K(\widehat{\boldsymbol{\eta}})\text{vec}\left(\widehat{\boldsymbol{\Gamma}}_X \cdot (\beta^0 - \widetilde{\beta}) + \boldsymbol{\mathcal{A}} - \widehat{\boldsymbol{\mathcal{A}}} + \boldsymbol{\varepsilon}\right) \\
&= \beta^0 + \widehat{\Omega}_X^{-1}B + \widehat{\Omega}_X^{-1}\text{vec}_K(\widehat{\boldsymbol{\eta}})'\text{vec}(\boldsymbol{\varepsilon}),
\end{aligned} \tag{13}$$

where $\widetilde{\beta}$, and $\widehat{\boldsymbol{\mathcal{A}}}$ are preliminary estimates of β^0, and $\boldsymbol{\mathcal{A}}$, respectively, used to form the estimate $\widehat{\boldsymbol{\Gamma}}_Y$. B is a bias term that must be sufficiently small order for standard asymptotic normality arguments. Specifically, $\widehat{\Omega}_X^{-1}B = o_p\left(\prod_{n=1} N_n^{-1/2}\right)$ is required. Hence, preliminary convergence rates are required on $(\boldsymbol{\Gamma}_X - \widehat{\boldsymbol{\Gamma}}_X)$, $(\beta^0 - \widetilde{\beta})$, and $(\boldsymbol{\mathcal{A}} - \widehat{\boldsymbol{\mathcal{A}}})$ to ensure bias is bounded appropriately. From Section 3.2, and the proof in Appendix A, with the kernel weighted estimator, $(\beta^0 - \widetilde{\beta})$ can be bounded by $O_p(\prod_n h_n)$ if $N_n^{-1/2} \lesssim h_n$, where h_n is the bandwidth used for dimension n. Likewise, $\left(\prod_{n=1} N_n^{-1}\right)\left(\text{vec}(\boldsymbol{\mathcal{A}} - \widehat{\boldsymbol{\mathcal{A}}})'\text{vec}(\boldsymbol{\mathcal{A}} - \widehat{\boldsymbol{\mathcal{A}}})\right)^{1/2} = O_p(\prod_n h_n)$. This leaves the bound for $(\boldsymbol{\Gamma}_X - \widehat{\boldsymbol{\Gamma}}_X)$, which is incidentally also useful to bound $\text{vec}_K(\widehat{\boldsymbol{\eta}})$.

Split the bias term, $B = B_1 + B_2$, and let $N := \prod_{n=1} N_n$. Then,

$$\begin{aligned}
B_1 &:= \text{vec}_K(\widehat{\boldsymbol{\eta}})'\text{vec}(\widehat{\boldsymbol{\Gamma}}_X) \cdot (\beta^0 - \widetilde{\beta}) \\
B_2 &:= \text{vec}_K(\widehat{\boldsymbol{\eta}})'\text{vec}(\boldsymbol{\mathcal{A}} - \widehat{\boldsymbol{\mathcal{A}}}).
\end{aligned}$$

First, $\text{vec}_K(\widehat{\boldsymbol{\eta}})$ appears in both, which can be split into $\text{vec}_K(\widehat{\boldsymbol{\eta}}) = \text{vec}_K(\boldsymbol{\Gamma}_X - \widehat{\boldsymbol{\Gamma}}_X) + \text{vec}_K(\boldsymbol{\eta})$. Let $\frac{1}{N}\text{vec}_K(\boldsymbol{\Gamma}_X - \widehat{\boldsymbol{\Gamma}}_X)'\text{vec}_K(\boldsymbol{\Gamma}_X - \widehat{\boldsymbol{\Gamma}}_X) = O_p(\xi_X^2)\mathbb{1}_K$, where ξ_X is the rate of consistency for the estimator of $\widehat{\boldsymbol{\Gamma}}_X$. For the two terms, take $(\beta^0 - \widetilde{\beta}) = O_p(\xi_\beta)$ and $\frac{1}{N}\text{vec}(\boldsymbol{\mathcal{A}} - \widehat{\boldsymbol{\mathcal{A}}})'\text{vec}(\boldsymbol{\mathcal{A}} - \widehat{\boldsymbol{\mathcal{A}}}) = O_p(\xi_{\mathcal{A}}^2)$, where these rates can be taken from Proposition 2. Then,

$$\begin{aligned}
\frac{1}{N}B_1 &= \frac{1}{N}\text{vec}_K(\boldsymbol{\Gamma}_X - \widehat{\boldsymbol{\Gamma}}_X)'\text{vec}(\widehat{\boldsymbol{\Gamma}}_X) \cdot (\beta^0 - \widetilde{\beta}) + \frac{1}{N}\text{vec}_K(\boldsymbol{\eta})'\text{vec}(\widehat{\boldsymbol{\Gamma}}_X) \cdot (\beta^0 - \widetilde{\beta}) \\
&= O_p(\xi_X\xi_\beta)\iota_K \sum_{k=1}^{K} \frac{1}{\sqrt{N}} \left\|\widehat{\boldsymbol{\Gamma}}_{X_k}\right\|_F + \frac{1}{N}\sum_{k=1}^{K} \text{vec}_K(\boldsymbol{\eta})'\text{vec}(\widehat{\boldsymbol{\Gamma}}_{X_k}) \cdot O_p(\xi_\beta).
\end{aligned}$$

Then, if $\boldsymbol{\eta}_{k,i_1\dots i_d} \perp\!\!\!\perp \widehat{\boldsymbol{\Gamma}}_{X_{k'},i_1\dots i_d}$ for all $k,k'=1,\dots K$, and $\frac{1}{\sqrt{N}}\left\|\widehat{\boldsymbol{\Gamma}}_{X,k}\right\|_F$ is bounded,

$$\frac{1}{N}B_1 = O_p(\xi_X\xi_\beta)\iota_K + O_p(\xi_\beta N^{-1/2})\iota_K$$

Likewise, for B_2,

$$\frac{1}{N}B_2 = O_p(\xi_X\xi_{\mathcal{A}})\iota_K + \frac{1}{N}\mathrm{vec}_K(\boldsymbol{\eta})'\mathrm{vec}(\boldsymbol{\mathcal{A}} - \widehat{\boldsymbol{\mathcal{A}}}),$$

such that if $\boldsymbol{\eta}_{,i_1\dots i_d} \perp\!\!\!\perp \widehat{\boldsymbol{\mathcal{A}}}_{i_1\dots i_d}$

$$\frac{1}{N}B_2 = O_p(\xi_X\xi_{\mathcal{A}})\iota_K + B_{21},$$

where $\sqrt{N}B_{21}$ converges to mean zero, variance zero normal distribution.

Under a full rank assumption on the transformed covariates, e.g. Assumption 7, then $\widehat{\Omega}_X^{-1} = O_p(1/N)\mathbb{1}_{K\times K}$, such that,

$$\sqrt{N}(\widehat{\beta}_I - \beta^0) = \sqrt{N}\left(O_p(\xi_X\xi_\beta)\iota_K + O_p(\xi_X\xi_{\mathcal{A}})\iota_K + O_p(\xi_\beta N^{-1/2})\iota_K + \widehat{\Omega}_X^{-1}\mathrm{vec}_K(\widehat{\boldsymbol{\eta}})'\mathrm{vec}(\boldsymbol{\varepsilon})\right).$$

If $\xi_X\xi_\beta = o_p(N^{-1/2})$ and $\xi_X\xi_{\mathcal{A}} = o_p(N^{-1/2})$, asymptotic bias is of order $o_p(N^{-1/2})$ and under certain conditions asymptotic normality can be established for the final term. Proposition 2 and its proof establish that the upper bound on the convergence rate for ξ_β and $\xi_{\mathcal{A}}$ can be $O_p(N^{-1/2})$, such that for any $\xi_X = o_p(1)$, the $o_p(N^{-1/2})$ convergence rate for asymptotic bias is achievable.

Dependence between either $\boldsymbol{\eta}$, respectively $\boldsymbol{\varepsilon}$, and $\widehat{\boldsymbol{\Gamma}}_X$ or $\widehat{\boldsymbol{\mathcal{A}}}$ can occur because the estimator $\widehat{\boldsymbol{\Gamma}}_X$, and/or $\widehat{\boldsymbol{\mathcal{A}}}$, may be functions of $\boldsymbol{\eta}$, respectively $\boldsymbol{\varepsilon}$. To break this dependence, a simple sample splitting procedure can be implemented. Say the data is split into two sections. The first section can be used for estimation of $\widehat{\boldsymbol{\Gamma}}_X$ and $\widehat{\boldsymbol{\mathcal{A}}}$, and the second section can be used for estimation of $\widehat{\beta}_I$. Hence, as long as $\boldsymbol{\varepsilon}$ and $\boldsymbol{\eta}$ are suitably independent across observations, the dependence introduced by the estimating equations is broken. A concrete example of a sample splitting procedure for panel data is outlined in Freeman and Weidner (2022), which is easily transferable to this setting.

Assumption 8. *$\{\varepsilon_{i_1\dots i_d}, \eta_{i_1\dots i_d}\}$, are i.i.d. across $i_1\dots i_d$.*

Assumption 9. $\mathbb{E}\big(\varepsilon^2_{i_1\dots i_d}|\eta_{i_1\dots i_d}\big) = \mathbb{E}\big(\varepsilon^2_{i_1\dots i_d}\big) =: \sigma^2_\varepsilon \le M < \infty$

Assumption 10. *$\mathbb{E}\left[\eta_{i_1\dots i_d}\eta'_{i_1\dots i_d}\right]$ is non–singular and $E\left[\varepsilon_{i_1\dots i_d}|\eta_{i_1\dots i_d}\right] = 0$.*

Assumption 11. *Let ξ_X, ξ_β, and $\xi_{\mathcal{A}}$ be sequences bounded in probability as $N_n \to \infty$ for each $n = 1,\dots d$. Maintain $N := \prod_{n=1}^d N_n$.*

(i). $\left(\left\|\widehat{\boldsymbol{\Gamma}}_{X_k} - \boldsymbol{\Gamma}_{X_k}\right\|_F^2/N\right)^{1/2} = O_P(\xi_X)$ *for each $k = 1,\dots K$,*

(ii). $\left(\|\widehat{\boldsymbol{\mathcal{A}}}-\boldsymbol{\mathcal{A}}\|_F^2/N\right)^{1/2}=O_P\left(\xi_{\mathcal{A}}\right)$,

(iii). $\|\beta-\widetilde{\beta}\|=O_p\left(\xi_\beta\right)$.

As $N_n\to\infty$ *for each* $n=1,\dots d$, $\xi_X\to 0$, $\xi_X\xi_\beta=o_p(N^{-1/2})$ *and* $\xi_X\xi_{\mathcal{A}}=o_p(N^{-1/2})$.

Assumption 11.(i) implies that there exists a consistent estimate of $\mathbb{E}\left(\boldsymbol{X}|\boldsymbol{\mathcal{A}}\right)$, with an arbitrary convergence rate. For implementation, the kernel weighted or group fixed-effect estimator, assuming X follows a pure interactive fixed-effect model can achieve this with regularity conditions on how $\boldsymbol{\mathcal{A}}$ enters the data generating process for X. Note, the convergence rate ξ_X can be arbitrarily slow as long as $\xi_{\mathcal{A}}$ and ξ_β converge to zero sufficiently fast. Hence, in the context of Proposition 2, this is not an overly strong assumption on $\widehat{\boldsymbol{\Gamma}}_{X_k}$. The convergence rates in Assumption 11.(ii)-(iii) can be taken directly from Proposition 2, e.g., $\xi_{\mathcal{A}}=O_p(1/\sqrt{N})$, and $\xi_\beta=O_p(1/\sqrt{N})$ under some additional regularity conditions noted below.

Note also that Assumption 11 details a potential reprieve from $\xi_{\mathcal{A}}=O_p(1/\sqrt{N})$, and $\xi_\beta=O_p(1/\sqrt{N})$ if indeed it is possible for ξ_X to converge at a faster rate. For example, to achieve $\xi_{\mathcal{A}}=O_p(1/\sqrt{N})$, and $\xi_\beta=O_p(1/\sqrt{N})$, it is necessary that the bandwidth parameter h_n shrink at the rate $1/\sqrt{N_n}$ for each $n=1,\dots,d$, faster than $O_p({N_n}^{-2/5})$ usually imposed for nonparametric estimation.[4] Hence, for sufficiently fast convergence rate ξ_X, the bandwidth parameter h_n has slack, and can converge slower for better efficiency.

The next assumption restricts $\boldsymbol{\eta}$ and $\boldsymbol{\varepsilon}$ to be independent of $\widehat{\boldsymbol{\Gamma}}_X$ and $\widehat{\boldsymbol{\mathcal{A}}}$. As mentioned already, a sample splitting device can be used to achieve this independence, given an appropriate independence condition such as Assumption 8.

Assumption 12. *Let* $\widehat{\boldsymbol{\Gamma}}_X$ *be the estimate for* $\boldsymbol{\Gamma}_X$ *and* $\widehat{\boldsymbol{\mathcal{A}}}$ *the estimate for* $\boldsymbol{\mathcal{A}}$ *in* (12). *Then,*

(i). $\boldsymbol{\eta}_{k,i_1\dots i_d}\perp\!\!\!\perp\boldsymbol{\mathcal{A}}_{i_1\dots i_d},\widehat{\boldsymbol{\mathcal{A}}}_{i_1\dots i_d}$ *for all* $k=1,\dots K$.

(ii). $\boldsymbol{\eta}_{k,i_1\dots i_d}\perp\!\!\!\perp\widehat{\boldsymbol{\Gamma}}_{X_{k'},i_1\dots i_d}$ *for all* $k,k'=1,\dots K$.

(iii). $\boldsymbol{\varepsilon}_{i_1\dots i_d}\perp\!\!\!\perp\widehat{\boldsymbol{\Gamma}}_{X'_k,i_1\dots i_d}$ *for all* $k=1,\dots K$.

The implications of Assumption 12.(i)-(ii) on the convergence of the inference estimator are already discussed above. Independence between $\boldsymbol{\eta}_{k,i_1\dots i_d}$ and $\boldsymbol{\mathcal{A}}_{i_1\dots i_d}$ in Assumption 12.(i) is made for convenience for a cleaner expression of the term B_{21} above, but can be relaxed under further regularity conditions on the moments of $\boldsymbol{\eta}_{k,i_1\dots i_d}$. Assumption 12.(iii) implies that in (13), $\widehat{\Omega}_X^{-1}\mathrm{vec}_K(\widehat{\boldsymbol{\eta}})'\mathrm{vec}(\boldsymbol{\varepsilon})=\widehat{\Omega}_X^{-1}\mathrm{vec}_K(\boldsymbol{\eta})'\mathrm{vec}(\boldsymbol{\varepsilon})+o_p(N^{-1/2})$. Under the assumptions imposed

[4]This required bandwidth rate is under Assumption6, and the required bandwidth rate is even stronger under the weaker requirement in AssumptionA.1 in the Appendix.

in Proposition 2, with $C_n^{-1} \asymp h_n = O_p(1/\sqrt{N_n})$ for all $n = 1, \dots, d$ and $\mathcal{M} = \{1, \dots, d\}$, and Assumptions 8-12, maintaining that $N := \prod_{n=1} N_n$,

$$\begin{aligned}\sqrt{N}(\widehat{\beta}_I - \beta^0) &= o_p(1) + \left(\frac{1}{N}\widehat{\Omega}_X\right)^{-1} \frac{1}{\sqrt{N}}\text{vec}_K(\boldsymbol{\eta})'\text{vec}(\boldsymbol{\varepsilon}) \\ &= o_p(1) + \left(\mathbb{E}\left[\boldsymbol{\eta}_{i_1\dots i_d}\boldsymbol{\eta}'_{i_1\dots i_d}\right] + o_p(1)\right)^{-1} \frac{1}{\sqrt{N}}\text{vec}_K(\boldsymbol{\eta})'\text{vec}(\boldsymbol{\varepsilon})\end{aligned}$$

Under a central limit theorem for the term $\frac{1}{\sqrt{N}}\text{vec}_K(\boldsymbol{\eta})'\text{vec}(\boldsymbol{\varepsilon})$, the estimator is then asymptotically normal with variance,

$$\mathbb{E}\left[\boldsymbol{\eta}_{i_1\dots i_d}\boldsymbol{\eta}'_{i_1\dots i_d}\right]^{-1} \Omega \mathbb{E}\left[\boldsymbol{\eta}_{i_1\dots i_d}\boldsymbol{\eta}'_{i_1\dots i_d}\right]^{-1}.$$

Under Assumption 8 and 9, $\Omega = \mathbb{E}\left[\boldsymbol{\eta}_{,i_1\dots i_d}\boldsymbol{\eta}'_{,i_1\dots i_d}\right]\sigma_\varepsilon^2$, for $\sigma_\varepsilon^2 := \mathbb{E}(\boldsymbol{\varepsilon}^2_{i_1\dots i_d})$, such that the variance simplifies in the standard way to $\sigma_\varepsilon^2\mathbb{E}\left[\boldsymbol{\eta}_{,i_1\dots i_d}\boldsymbol{\eta}'_{,i_1\dots i_d}\right]^{-1}$. Therefore,

$$\sqrt{N}\left(\widehat{\beta}_I - \beta^0\right) \xrightarrow[d]{} \mathcal{N}\left(0, \sigma_\varepsilon^2\mathbb{E}\left[\boldsymbol{\eta}_{i_1\dots i_d}\boldsymbol{\eta}'_{i_1\dots i_d}\right]^{-1}\right)$$

Some additional regularity conditions are required to make sure estimates of the fixed-effects that are used as proxies in the kernel weights converge at the sufficient rate.

Assumption 13 (Bai (2009) conditions)**.** *(i).* $\mathbb{E}\|X_{i_1\dots i_d}\|^4$ *bounded*

(ii). For each $n = 1, \dots, d$, $\mathbb{E}\|\varphi^{(n)}_{i_n}\|^4$ *bounded, and* $\frac{1}{N_n}\sum_{i_n=1}^{N_n}\varphi^{(n)}_{i_n}\varphi^{(n)\prime}_{i_n}$ *converges to an* $r_n \times r_n$ *p.d. matrix as* $N_n \to \infty$

(iii). $\mathbb{E}\varepsilon_{i_1\dots i_d} = 0$, $\mathbb{E}(\varepsilon_{i_1\dots i_d})^8$ *bounded*

(iv). $\varepsilon_{i_1\dots i_d}$ *independent of* $X_{i'_1\dots i'_d}$ *and* $\varphi^{(n)}_{i'_n}$ *for all* $i_1\dots i_d$ *and* $i'_1\dots i'_d$, *and* $n = 1, \dots, d$.

Corollary 1. *If Assumptions 8 and 13 hold, then* $C_n^{-2} = 1/N_n$ *for each* $n = 1, \dots, d$ *if the interactive fixed-effect estimator from Bai (2009) is used in conjunction with the matrix low-rank approximation estimator from Section 3.1 separately for each* $n = 1, \dots, d$.[5]

Theorem 1 (Asymptotic distribution under homoskedasticity)**.** *Let the assumptions in Proposition 2 hold. Additionally, let Assumptions 8-13 hold. Then,*

$$\sqrt{N}\left(\widehat{\beta}_I - \beta^0\right) \xrightarrow[d]{} \mathcal{N}\left(0, \sigma_\varepsilon^2\mathbb{E}\left[\boldsymbol{\eta}_{i_1\dots i_d}\boldsymbol{\eta}'_{i_1\dots i_d}\right]^{-1}\right)$$

Consistent estimation of the variance term is possible under the same set of assumptions. Estimate $\widehat{\boldsymbol{\varepsilon}} = \boldsymbol{Y} - \boldsymbol{X} \cdot \widehat{\beta}_I - \widehat{\boldsymbol{\mathcal{A}}}$. Then, $\frac{1}{N}\sum_{i_1\dots i_d}\widehat{\varepsilon}^2_{i_1\dots i_d} = \sigma_\varepsilon^2 + o_p(1)$ follows from the consistency of $\widehat{\boldsymbol{\Gamma}}_Y$, $\widehat{\boldsymbol{\Gamma}}_X$, and $\widehat{\beta}_I$. Likewise $\mathbb{E}\left[\boldsymbol{\eta}_{i_1\dots i_d}\boldsymbol{\eta}'_{i_1\dots i_d}\right]$ can be estimated consistency from the sample analog

[5] See Proposition A.1 in Bai (2009) appendix

$\frac{1}{N}\sum_{i_1\dots i_d}\left(\boldsymbol{X}_{i_1\dots i_d}-\widehat{\boldsymbol{\Gamma}}_{X,i_1\dots i_d}\right)\left(\boldsymbol{X}_{i_1\dots i_d}-\widehat{\boldsymbol{\Gamma}}_{X,i_1\dots i_d}\right)'$. Because of the weighted-within transformations, some degrees of freedom adjustment is required to calculate the standard errors in finite samples.

Some less restrictive assumptions on the error term can be used, and would require Lyapunov conditions. Alternatively, a central limit theorem can be assumed, as is common in the interactive fixed-effects literature, for cases that drop the identically distributed assumption.[6]

Assumption 14. *$\{\varepsilon_{i_1\dots i_d}, \eta_{i_1\dots i_d}\}$, are independent across $i_1\dots i_d$.*

Assumption 15. *(i). $\sigma^2_{i_1\dots i_d} := \mathbb{E}\big(\varepsilon^2_{i_1\dots i_d}|\boldsymbol{\eta}\big)$ is bounded.*

(ii). For nonrandom positive definite Σ,

$$\operatorname*{plim}_{N\to\infty}\frac{1}{N}\sum_{i_1\dots i_d}\sigma^2_{i_1\dots i_d}\eta_{i_1\dots i_d}\eta'_{i_1\dots i_d}=\Sigma,$$

$$\frac{1}{\sqrt{N}}\sum_{i_1\dots i_d}\eta_{i_1\dots i_d}\varepsilon_{i_1\dots i_d}\xrightarrow{d}\mathcal{N}(0,\Sigma).$$

Assumption 15 can be amended to restrict how much heteroskedasticity and correlation across units is allowed. As it is stated, Assumption 15 allows for very general error second moments. Then,

$$\sqrt{N}\big(\widehat{\beta}_I-\beta^0\big)\xrightarrow{d}\mathcal{N}(0,\Omega_X^{-1}\,\Sigma\,\Omega_X^{-1}),\qquad \Omega_X:=\operatorname*{plim}_{N\to\infty}\frac{1}{N}\sum_{i_1\dots i_d}\eta_{i_1\dots i_d}\eta'_{i_1\dots i_d}.$$

Theorem 2 (Asymptotic distribution under heteroskedasticity)**.** *Let the assumptions in Proposition 2 hold. Additionally, let Assumptions 9-15 hold. Then,*

$$\sqrt{N}\big(\widehat{\beta}_I-\beta^0\big)\underset{d}{\to}\mathcal{N}\left(0,\Omega_X^{-1}\,\Sigma\,\Omega_X^{-1}\right).$$

For simulated coverage, standard errors are estimated with an estimated degree of freedom adjustment. The degree of freedom adjustment significantly improves finite sample coverage, as the standard errors can be under-estimated if the projection from the weighted-within transformation is not accounted for. For implementation, the parametric bootstrap was also implemented to estimate the standard errors. Simulation exercises suggest the bootstrap methods offer overly-conservative coverage of the true parameter value, with standard errors much larger than necessary for nominal coverage.

[6] See Assumption E in Section 5 in Bai (2009) for a leading example of assuming a central limit theorem in this setting.

4 Discussion of estimators

This section serves to discuss the results in Section 3, motivate further some of the chosen methods, and provide some methods to estimate cluster assignments or proxies for kernel weights.

4.1 Matrix method results

As stated already, Proposition 1 takes for granted the dimension to flatten across admits a low rank interactive fixed-effect term for the least square method in Bai (2009). Under Assumption 1.(ii) the singular values of the flattened normalised noise term dissipates as follows;

$$\frac{1}{\sqrt{\prod_n N_n}} \left\| \varepsilon_{(n)} \right\| = O_p \left(\frac{1}{\sqrt{\min\{N_n, \prod_{m \neq n} N_m\}}} \right).$$

$\mathcal{A}$ is commonly regularised such that the normalised singular values of its flattenings are $O_p(1)$, that is, the singular values are not asymptotically negligible like those of the noise term. This ensures that, after flattening $\mathcal{A}$, each of the singular values eventually dominate those of the noise term. These conditions make up similar restrictions imposed in Ahn and Horenstein (2013) that allow for the use of the eigenvalue ratio test (ER) to diagnose the number of factors. However, this test is stipulated in a pure factor model setting, so its use is limited here.

Consider the factor model applied to a flattening that may not be low-rank. For a concrete example of when this can occur see the data generating process in the simulations in Section 5, where $\varphi^{(n)}$ are designed to be low-dimensional for some n, and high-dimensional otherwise. Along the dimensions of $\mathcal{A}$ that do not conform to the low-rank assumption in Assumption 3, the tail singular values may become difficult to discern from the singular values of the noise term in small samples. This means variation from those tail factors are less likely to be projected out from the factor model unless many factors are used in this projection. If r_n for $n \notin \mathcal{L}$ is allowed to increase adversely, for example at exactly the upper bound, then factor projection may never sufficiently project all relevant factors. Also, as the number of estimated factors increases, Assumption 4 becomes harder to satisfy since variation in the set of covariates is also projected out. This demonstrates the importance of choosing the correct dimension to flatten over, which is supported by the simulation results in Section 5.

Hence, a standard factor model that estimates at least r_n factors should result in consistent estimation of the slope coefficients, see Moon and Weidner (2015). However, this relies on an important structural feature of the unobserved heterogeneity term. When flattened in the chosen dimension – the first dimension in the above example – the rank of the matrix after flattening must be low relative to data size. This implies that to successfully project out the variation in the fixed-effect term either the matrix of fixed-effects from any flattening is low-rank, or, at least one flattening leads to a low-rank matrix of fixed-effects and the analyst knows which flattening this is. To use the above example again, this means the analyst knows that $\varphi^{(1)}\Gamma_1'$ is low-rank, hence flattening in the first dimension is the correct way to recast the model to a panel data model, and so forth for the other flattenings. Whilst requiring low-rankness in at least one dimension may be an acceptable restriction, having knowledge of which dimension this low-rankness resides in is potentially more restrictive.

To understand the problem, consider the following two examples, one where the flattening is not low-rank and one where it is. First, assume $\varphi^{(1)}$ varies in a high-dimensional parameter space, e.g. with $N_1 < N_2N_3$, $\varphi^{(1)} \in \mathbb{R}^{N_1 \times N_1}$ and $\Gamma \in \mathbb{R}^{N_2N_3 \times N_1}$ with each column mutually orthogonal for both these matrices. Then the product of these matrices is full-rank and any factor projection approach will not fully control for this term. On the contrary, consider $\varphi^{(1)} \in \mathbb{R}^{N_1 \times N_1}$ where all columns are linearly dependent. Then the matrix $\varphi^{(1)}\Gamma'$ is rank-1 regardless of L and of how $\varphi^{(2)}$ and $\varphi^{(3)}$ vary, thus can be projected with a factor model estimated with 1 factor. Hence it is important which dimension the analyst chooses to flatten over. This exact situation is exemplified in simulations in Section 5.

Well established diagnostics in Bai and Ng (2002), Ahn and Horenstein (2013) and Hallin and Liška (2007) can be used to determine the number of factors in the pure factor model. These diagnostics can be repeated across different flattenings, which may be informative of the dimension to use for flattening, although no formal results are provided for using these. In the standard panel model setting, establishing the number of terms in the interactive fixed-effect in the presence of covariates is still an active area of research.

4.2 Estimating kernel weight proxies

Discussed here are some functionals of multidimensional arrays that are useful for estimating proxies to formulate kernel weights. This includes a discussion on how to uncover proxies from multidimensional array data, and a discussion on why standard matrix methods do not extend well to the multidimensional setting.

For the weighted-within estimator, it is important to find proxies that isolate variation in each dimension since weights are formed separately for each index. Discussed here are decompositions of multidimensional arrays that can perform this, Kolda and Bader (2009) contains a nice summary of some candidate decompositions. The method discussed here uses the higher order singular value decomposition (HOSVD), and focuses on components of this decomposition that have well formulated theoretical properties. The HOSVD is traditionally used in pursuit of a low-rank tensor decomposition by either direct truncation of left singular vectors or by some iteration approach similar to this, see for example the higher order orthogonal iteration scheme. The problem of direct truncation, however, is not well-posed because the solution to the low-rank tensor problem may not be unique and reformulating the original tensor after the aforementioned truncation is not guaranteed to be lower tensor rank. See De Silva and Lim (2008) for an extensive explanation of the ill-posedness issues. Hence, this method cannot be used in the pursuit of analytic consistency results. Problems also arise in this setting where the reformulated tensors can be arbitrarily well approximated by a tensor of lower tensor rank, which is a result of the border rank issue of the tensor rank decomposition.

Reconsider the three dimensional model with heterogeneity of the following form

$$\mathcal{A}_{ijt} = \sum_{\ell=1}^{L} \varphi_{i\ell}^{(1)} \varphi_{j\ell}^{(2)} \varphi_{t\ell}^{(3)}. \tag{14}$$

As shown in De Silva and Lim (2008), the Eckart–Young–Mirsky theorem cannot be relied upon to guarantee an optimal low-rank approximation for the multidimensional array $\boldsymbol{\mathcal{A}}$. Also reconsider the singular value decomposition for matrices, applied to each of the n-flattenings of $\boldsymbol{\mathcal{A}} \in \mathbb{R}^{N_1 \times, \ldots, \times N_d}$ as

$$\mathcal{A}_{(n)} = U^{(n)} \Sigma_n V^{(n)\prime}. \tag{15}$$

By the same logic as in the matrix case and formalised with the Eckart-Young-Mirsky theorem, variation over the rows of each $U^{(n)}$ explains variation over the n^{th} dimension of the multidimensional array $\boldsymbol{\mathcal{A}}$. Thus, if $\mathcal{A}_{(n)}$ is low rank, the leading few columns of $U^{(n)}$ provide good proxies for closeness in n^{th} dimension of $\boldsymbol{\mathcal{A}}$. If $\mathcal{A}_{(n)}$ is not low rank then these leading columns still provide good proxies for bias reduction using the kernel-weighted fixed-effects. Hence, by reconsidering the tensor problem as a sequence of matrix problems, the usual singular value decomposition properties can be utilised for this bias reduction.

Consider for any of the dimensions n the corresponding matrix of left singular vectors from above, $U^{(n)}$, estimated with noise ε_{ijt}. That is, each $U^{(n)}$ are calculated from the matrix $\boldsymbol{\mathcal{V}}_{(n)} = \boldsymbol{\mathcal{A}}_{(n)} + \boldsymbol{\varepsilon}_{(n)}$. Under regularity conditions on the noise term $\boldsymbol{\varepsilon}$, the left singular vectors from this decomposition consistently estimate $U^{(n)}$, up-to-scalings and rotations. For example, in the three dimensional case, define the r_1-vector $\widehat{U}_i^{(1)}$ as the i-th row of the left singular matrix of $\boldsymbol{\mathcal{A}} + \boldsymbol{\varepsilon}$ flattened in the first dimension. Then the vector $\widehat{U}_i^{(1)}$ may comprise of,

$$\widehat{U}_i^{(1)} = \varphi_i^{(1)} + O_p\left(\frac{1}{\sqrt{\min\{N_1, N_2 N_3\}}}\right).$$

Likewise, for any dimension n define $\widehat{U}^{(n)}$ as the matrix of singular vectors from $\boldsymbol{\mathcal{A}} + \boldsymbol{\varepsilon}$ flattened in the n-th dimension, where $\boldsymbol{\mathcal{A}}$ is the unobserved fixed-effects component of interest and $\boldsymbol{\varepsilon}$ is the usual idiosyncratic noise term. Then, Bai and Ng (2002) detail conditions required for the following "up-to-rotation" consistency result, which has been amended to this paper's setting;

Lemma 1 (Theorem 1 from Bai and Ng (2002))**.** *For any fixed integer $k \geq 1$, there exists an $(r_n \times k)$ matrix H_n^k with $\text{rank}(H_n^k) = \min\{k, r_n\}$ and $C_n = \min\left\{\sqrt{N_n}, \prod_{n' \neq n} \sqrt{N_{n'}}\right\}$ such that for each n under some regularity conditions*

$$C_n^2 \left\| \widehat{U}_{i_n}^{(n)} - H_n^{k\prime} \varphi_{i_n}^{(n)} \right\|^2 = O_p(1).$$

This establishes a consistency result for estimating weight proxies and suggests these left singular vectors are viable options to input into kernels to formulate weights in each dimension if the true error, $\boldsymbol{\mathcal{V}} = \boldsymbol{\mathcal{A}} + \boldsymbol{\varepsilon}$, is observed. It also makes concrete the limitation implied by the value of C_n for each index - that short indices have poorly estimated proxies. Given that the error term displayed in (**??**) is multiplicative across dimension, the error from this poor approximation should become negligible as long as enough other dimension proxies are well estimated. Also, the presence of the rotation matrices, H_n^k, in Lemma 1 can be ignored since these do not change relative distances of each unit under standard distance metrics used in the kernel weights.

However, it is not necessarily justified to assume the true error, $\boldsymbol{\mathcal{V}} = \boldsymbol{\mathcal{A}} + \boldsymbol{\varepsilon}$, is observed. In fact, an estimate of the error $\widehat{\mathcal{V}}_{ijt} = Y_{ijt} - X_{ijt}\widehat{\beta} = X_{ijt}(\beta^0 - \widehat{\beta}) + \mathcal{A}_{ijt} + \varepsilon_{ijt}$, clearly depends on the estimate $\widehat{\beta}$, hence should be bound by the rate of convergence for this estimate. For this reason, the uniform convergence result from Lemma 1 is unlikely to be useful. Proposition A.1 in Bai (2009) does provide bounds for the mean squared deviation,

$$\frac{1}{N_n} \sum_{i_n=1}^{N_n} \left\| \widehat{U}_{i_n}^{(n)} - H_n^{k\prime} \varphi_{i_n}^{(n)} \right\|^2 = O_p(\|\beta^0 - \widehat{\beta}\|^2) + O_p\left(\frac{1}{\min\{N_n, \prod_{n' \neq n} N_{n'}\}} \right),$$

if $\widehat{U}_{i_n}^{(n)}$ come from the Bai (2009) least squares problem in each dimension. This establishes, ignoring H_n^k, $\frac{1}{N_n} \sum_{i_n=1}^{N_n} \left\| \widehat{U}_{i_n}^{(n)} - \varphi_{i_n}^{(n)} \right\|^2 = O_p\left(\frac{1}{N_n}\right)$ if each dimension sample size grows at a roughly comparable rate and the convergence rate for $\|\beta^0 - \widehat{\beta}\|$ in Proposition 1 is used.[7] Then, C_n^{-2} from Proposition 2 is $1/N_n$ and the parametric rate for the kernel weighted estimator can be established.

The invertible matrix H_n^k can be ignored in the above display because these preliminary estimates are used to form distance measure between units. Hence, since the true distance of interest is that between the true parameter values,

$$\left\| \varphi_{j_n}^{(n)} - \varphi_{i_n}^{(n)} \right\|^2 = \left\| (H_n^{k\prime})^{-1} \left(H_n^{k\prime} \varphi_{j_n}^{(n)} - H_n^{k\prime} \varphi_{i_n}^{(n)} \right) \right\|^2 \leq \frac{1}{\lambda_{\min}(H_n^{k\prime})} \left\| H_n^{k\prime} \varphi_{j_n}^{(n)} - H_n^{k\prime} \varphi_{i_n}^{(n)} \right\|^2$$

[7] To be precise, the comparable rate restriction for any dimension $n = 1, \ldots, d$ is $N_n \lesssim \prod_{n' \neq n} N_{n'}$. The comparable rate commonly used in the two-dimensional panel literature is $N \approx T$. Hence, in the multidimensional problem, the relative rate growth is milder than the two-dimensional case.

where the inequality is a spectral norm bound, and $\lambda_{\min}(H_n^{k\prime})$ is the smallest eigenvalue of $H_n^{k\prime}$, which is strictly positive from invertibility. Therefore, it is without loss of generality to bound the space of true parameters, or the space of true parameters once rotated by $H_n^{k\prime}$.

Notice, for the space of the proxy measures - $\widehat{U}_{i_n}^{(n)}$ in the above example - to span the space of the true fixed-effects $\varphi_{i_n}^{(n)}$, it is necessary that the product space of fixed-effects over $n' \neq n$ has rank greater than $\varphi^{(n)}$. This is not, however, problematic in the context of this paper. Take the example where the $\widehat{U}_{i_n}^{(n)}$ indeed spans a lower rank space than the ambient space of $\varphi_{i_n}^{(n)}$, but are still estimates of linear combinations of $\varphi_{i_n}^{(n)}$. This could happen in the situation where, for example, the space of $\varphi^{(n)}$ has dimension greater than the product space over $\varphi^{(n')}$ for $n' \neq n$. Then $\widehat{U}_{i_n}^{(n)}$ actually estimates $U_{i_n}^{(n)} = \widetilde{H}\varphi_{i_n}^{(n)}$, a linear combination where $\text{rank}(\widetilde{H}) = r_n < \text{rank}(\varphi^{(n)})$. The reason this is not problematic here is that the normed difference $\|\varphi_{i_n}^{(n)} - \varphi_{i_n'}^{(n)}\|$ can be bounded by $\|\widehat{U}_{i_n}^{(n)} - \widehat{U}_{i_n'}^{(n)}\|$ and some additional fixed-effect estimation errors. Hence, the results still carry through for the case where only lower-dimensional linear combinations of fixed-effects can be identified and estimated.

5 Simulation

Two simulation exercises are performed. The first analyses the performance of the matrix methods compared to the kernel weighted methods when N and T are allowed to increase. These results are summarised in Figure 1. The second simulation exercise analyses some fixed sample properties of the set of estimators when Assumption 3 is violated in some dimensions. This assumption governs the rank of the fixed-effect term after it is transformed into matrix form. These results are summarised in Tables 2-3.

5.1 Growing sample exercise

Data is generated as,

$$\begin{aligned} Y_{ijt} &= X_{ijt}\beta + \mathcal{A}_{ijt} + \varepsilon_{ijt} \\ X_{ijt} &= \mathcal{A}_{ijt} + \eta_{ijt} \end{aligned}$$

with $\mathcal{A}_{ijt} = \sum_{\ell=1}^{2} \varphi_{i\ell}^{(1)} \varphi_{j\ell}^{(2)} \varphi_{t\ell}^{(3)}$. Also,

$$\varepsilon_{ijt} \overset{i.i.d.}{\sim} N(0,1), \eta_{ijt} \overset{i.i.d.}{\sim} N(0, \rho\eta_{ijt}^2) \text{ and for each } \ell,\ \varphi_{i\ell}^{(1)}, \varphi_{j\ell}^{(2)}, \varphi_{t\ell}^{(3)} \overset{i.i.d.}{\sim} N(0,1).$$

The results depicted in Figure 1 show two specifications for the variance of $\mathcal{A}_{ijt}$. $\mathcal{A}_{ijt}$ is normalised to have low variance in the left panel, then high variance in the right panel. Estimators considered use 2 or 3 estimated factors such that the multilinear rank is exactly or over-predicted. The left panel shows that whilst the matrix methods, labelled Factor, converge to the true estimator slower than the empirical bounds, for the sample sizes considered this does not look egregious in the low variance case. However, inspection of the right panel of Figure 1 highlights that the finite sample bias, and apparent slow convergence rate for the matrix methods are exacerbated when variance for the fixed-effect term is increased. Indeed, for the sample size considered, the empirical bounds do not cover the true parameter value at a moderately low sample size.

This contrasts with the results for the kernel weighted fixed-effect projection. Finite sample bias is negligible compared to variance in both the left and right panel. This estimator has roughly the same variance as the matrix methods, but for the sample sizes considered, the empirical bounds are always valid.

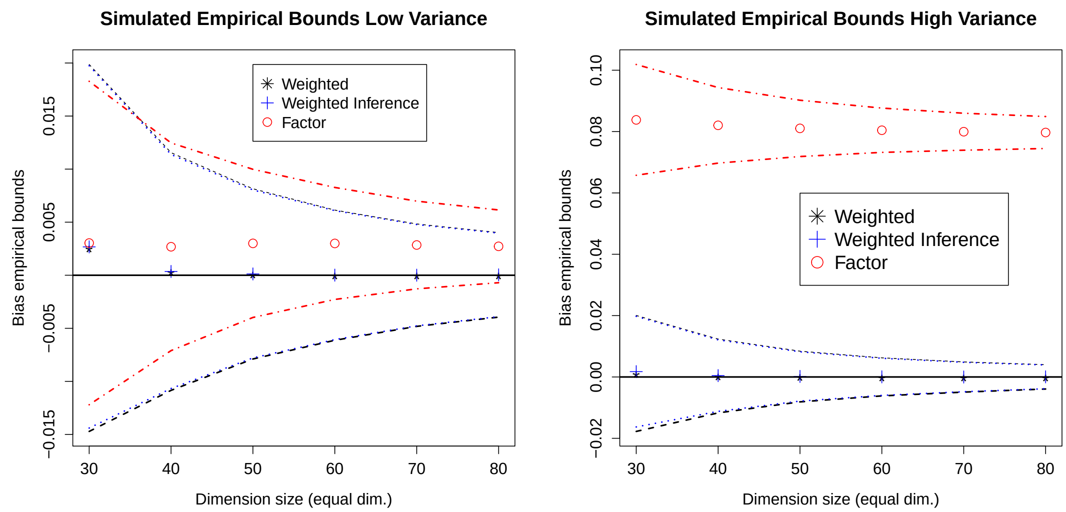

Figure 1: Mean bias of β estimates with 95% empirical bounds for growing sample size
Left panel, $sd(\varphi^{(n)}_{i_n\ell}) = 1$, right panel $sd(\varphi^{(n)}_{i_n\ell}) = 5^{1/3}$. 10,000 Monte Carlo rounds. Points depict mean bias. Dotted lines depict 95% empirical bounds. Factor depicts the matrix factor methods. Weighted and weighted inference depicts the kernel weighted method, and kernel weighted method with inference correction.

Table 1 displays coverage results and compares: (1), the kernel weighted estimator without vs. with the inference correction discussed in Section 3.3; and, (2), coverage when the multilinear rank is correct vs. over-predicted. For the first comparison, note that the Ker Inf coverage, which is coverage for the inference corrected estimator, is always better than the uncorrected estimator. This follows directly from the theoretical arguments made in Section 3.3. Second, note that over-predicting the multilinear rank improves coverage for the uncorrected estimator. This is reminiscent of the simulation results in traditional panel factor models, where in the presence of heteroskedasticity, over-prediction of rank can improve finite sample properties. The factor model undercovers in both the correct rank and over-predicted rank cases, with progressively worse coverage as the sample size grows.

	$\widehat{r}=2$	Coverage		$\widehat{r}=3$	Coverage	
Dim	Ker	Ker Inf	Factor	Ker	Ker Inf	Factor
30	0.86	0.96	0.90	0.91	0.94	0.89
40	0.87	0.96	0.88	0.93	0.95	0.88
50	0.86	0.96	0.85	0.93	0.95	0.82
60	0.86	0.95	0.80	0.93	0.95	0.76
70	0.85	0.94	0.75	0.93	0.95	0.70
80	0.85	0.95	0.68	0.93	0.95	0.61

Table 1: Estimated coverage for %5 nominal test

Ker stands for the kernel weighted estimator without inference correction. Ker Inf stands for the kernel weighted estimator with the inference correction. Factor is the matrix method used making the first dimension the rows, and the second and third dimensions jointly the columns. Dim stands for the size of each dimension, such that the total sample size is the cube of this. Coverage is measured for heteroskedasticity robust standard errors.

5.2 Fixed sample exercise

Table 2 shows simulation results for the following DGP,

$$\begin{aligned} Y_{ijt} &= X_{ijt}\beta + \mathcal{A}_{ijt} + \mathcal{B}_{ijt} + \varepsilon_{ijt} \\ X_{ijt} &= \mathcal{A}_{ijt} + \mathcal{B}_{ijt} + \eta_{ijt} \end{aligned}$$

with $\mathcal{A}_{ijt} = \sum_{\ell=1}^{N_1} \varphi_{i\ell}^{(1)} \varphi_{j\ell}^{(2)} \varphi_{t\ell}^{(3)}$, $\mathcal{B}_{ijt} = \alpha_{ij} + \gamma_{it} + \delta_{jt}$. Also,

$$\varepsilon_{ijt}, \eta_{ijt}, \alpha_{ij}, \gamma_{it} \text{ and } \delta_{jt} \overset{i.i.d.}{\sim} N(0,1) \text{ and for each } \ell,\ \varphi_{i\ell}^{(1)}, \varphi_{t\ell}^{(3)} \overset{i.i.d.}{\sim} N(0,1).$$
$$\varphi_{j1}^{(2)} \overset{i.i.d.}{\sim} N(0,1) \text{ with } \varphi_{j1}^{(2)} = \varphi_{j2}^{(2)} = \cdots = \varphi_{jN_1}^{(2)}$$

$\mathcal{A}_{ijt}$ and $\mathcal{B}_{ijt}$ are normalised to have unit variance. $\boldsymbol{\mathcal{A}}$ is specified such that it is rank 1 when flattened in the second dimension and rank N_1 when flattened in either dimension one or three. That is, the multilinear rank is $\mathbf{r} = (N_1, 1, N_1)$. This comes directly from the data generating process for each $\varphi^{(n)}$, where the matrix $\varphi^{(2)}$ is designed to be rank-1 and the matrices $\varphi^{(1)}$ and $\varphi^{(3)}$ are designed to be rank-N_1.

In Table 2, the estimators OLS and Fixed-effects are simply the pooled OLS estimator and the pooled OLS estimator after additive fixed-effects are projected out, respectively. As expected both of these two have poor bias. The factor model is used after first flattening along each dimension as Factor(dim $= n$), where n is the dimension used for flattening. In each case, 2 factors are projected. The results show the theoretical result succinctly, where the bias is close to zero when the correct dimension is flattened over (the second dimension in this case) and very poor bias when the incorrect dimension is used (the first and third dimensions). Lastly, the kernel differencing estimator is estimated with Gaussian kernel function with bandwidths 0.5, 1 and 1.5; which are standardised to be equivalent to standard deviations of the proxy measures. All kernel estimators have comparable bias but substantially better standard deviation for bandwidth equal 1 and 1.5.

This analysis is repeated for the four dimensional case in Table 2, where the second and third dimensions admit low-dimensional unobserved interactive fixed-effects parameters. The simulations suggest similar results as the three dimensional case, where the factor models perform well when flattened in the low-dimensional dimensions (second and third) and poorly in the high-dimensional dimensions (first and fourth).

3-D	Bias	St. dev.	RMSE	4-D	Bias	St. dev.	RMSE
OLS	0.6668	0.0033	0.6693		0.6670	0.0018	0.6683
Fixed-effects	0.4997	0.0114	0.5109		0.4981	0.0282	0.5256
Kernel (h = 0.5)	0.0030	0.0090	0.0949		5.23e-05	0.0114	0.1068
Kernel (h = 1.0)	0.0031	0.0068	0.0825		4.42e-05	0.0057	0.0755
Kernel (h = 1.5)	0.0037	0.0062	0.0788		-1.32e-05	0.0045	0.0671
Factor (dim = 1)	0.4319	0.0135	0.4473		0.3733	0.0311	0.4129
Factor (dim = 2)	0.0030	0.0050	0.0708		0.0030	0.0030	0.0549
Factor (dim = 3)	0.4319	0.0135	0.4473		0.0030	0.0030	0.0549
Factor (dim = 4)	NA	NA	NA		0.3734	0.0311	0.4129

Table 2: 3D model ($N_1 = N_2 = N_3 = 36$), with 10,000 Monte Carlo rounds.
4D model ($N_1 = N_2 = N_3 = N_4 = 20$), with 10,000 Monte Carlo rounds.
All results are in relation to β estimation.

A two-dimensional simulation exercise is also performed to compare the grouped and kernel weighted fixed-effects approaches to the factor model approach in a setting where theoretical results for the factor model are well known. Table 3 shows the results of this two-way setting where the data generating process is a factor model with two factors. No claim is made that any meaningful improvement can be made to the factor model estimator in the two dimensional setting, and it should be noted that the bias, while larger than the kernel estimators, is still small, and negligible compared to standard deviation.

	Mean bias	St. dev.	RMSE
OLS	0.6672	0.0033	0.6697
Fixed-effects	0.5000	0.0043	0.5042
Kernel (h = 0.5)	0.0003	0.0055	0.0742
Kernel (h = 1.0)	0.0003	0.0054	0.0735
Kernel (h = 1.5)	0.0004	0.0053	0.0728
Factor (R = 2)	0.0024	0.0047	0.0686
Factor (R = 4)	0.0031	0.0048	0.0694
Factor (R = 6)	0.0024	0.0049	0.0700

Table 3: 2D model ($N_1 = N_2 = 216$), with 10,000 Monte Carlo rounds.

6 Empirical application - demand estimation for beer

The methods proposed in this paper are applied to estimated the demand elasticity for beer. Price and quantity for beer sales is taken from the Dominick's supermarket dataset for the years 1991-1995 and is related to supermarkets across the Chicago area. Price and quantity vary over three dimensions in this example – product (i), store (j) and week, though these are aggregated to fortnights (t). Fixed-effects that interact across all three dimensions can control for temporal taste shocks to beer consumption that differ over both product and store. Take for instance a large sporting event (temporary t shock) that changes preferences differently across locations (j) and across certain subsets of sponsored beer (i). Analysts may want some way to control for such interactions over all observed dimensions, even if they do not explicitly observe such For example, in the stadiums for the many NBA finals playoffs the Chicago Bulls played in the early 1990's, Miller Lite beer advertisements could be seen alongside advertisements for a substitute product, Canadian Club whisky. This suggests these events attracted large marketing campaign spends for these and other beer substitute brands that most likely also included price offers at local supermarkets. Whilst the impact of these advertisements and price offers on the demand for or price of beer is not clear and, further, that it is reasonably safe to assume the econometrician does not observe the plethora of marketing campaigns around these events, the analyst would most likely still want to control for aggregate shocks like these.

Models for demand estimation ideally account for endogenous variation in prices and quantity. The classic instrumental variable approach is to find a variable that varies exogenously to the production process but can reasonably describe price fluctuations. A popular instrument in the estimation of beer demand is the commodity price for barley, one of the product's main ingredients, see e.g. Saleh (2014); Tremblay and Tremblay (1995); Richards and Rickard (2021).

Since the price of barley is arguably not driven by the demand for it by any one supplier of beer, it can be a useful variable to instrument for price shifts. In the following, it is taken as given that the price of barley is exogenous with respect to the fixed-effects and noise term, ε.

For validity, the instrument is also required to be strong, in the sense that it is strongly correlated with price. In this dataset, correlation between the price of barley, which varies over only t, and price of beer depends on how beer price is aggregated. If beer price is first integrated over i and j, such that it only varies over t, then it is highly correlated with the price of barley, at 0.61. However, if beer price is not aggregated at all it is only correlated at 0.05. This heuristic suggests there are important product and store level price drivers for beer that are not accounted for by fluctuations in the price of barley, or indeed by price fluctuations only over time. Price of barley is only observed at the fortnightly level, which further reduces variability after projecting beer prices onto this instrument. Standard errors for the IV estimator below are much larger than other estimators, which may be explained partly by this loss in effective sample size.[8]

Table 4 refers to the estimates for demand elasticities for the following regression model,

$$\log(quantity_{ijt}) = \log(price_{ijt})\beta + \mathcal{A}_{ijt} + \varepsilon_{ijt} \tag{16}$$

where $\mathcal{A}_{ijt}$ is the usual interactive fixed-effects term from the prequel. This amounts to estimating the standard log-log model for demand with fixed-effects. That is,

$$quantity_{ijt} = price_{ijt}^{\beta} \exp(\mathcal{A}_{ijt} + \varepsilon_{ijt}).$$

No additional controls are included here since they are low-dimensional and subsumed by the fixed-effects term. Estimates for pooled OLS are positive, a contradiction that demand curves are downward sloping. Whilst IV estimates are negative, standard errors are very large. As noted above, the effective sample size for the second stage of the instrumental variable approach is orders of magnitude smaller because the instrument only varies over time, specifically by fortnight. Estimates under the additive model for fixed-effects are also negative, with much smaller standard errors than the IV estimate.

The matrix methods, labelled Factor, are implemented in all three dimensions of the data. That is, the three-dimensional array is transformed into three different matrices, one with products as rows, one with stores as rows, and lastly with time as rows. In all three specifications, five factors are estimated. The results suggest the factor model for fixed-effects can be sensitive to how the array is organised into a two-dimension problem. Simulation results from Section 5 suggest a potential explanation - sparsity of the dimension specific fixed-effects in the interaction term can be heterogeneous, and imply a different factor model rank when the tensor of data is flattened, or matricised, in different dimensions. This in turn implies completely different

[8] The effective sample size for the IV estimator drops from $N_1 N_2 T$ to just T.

models when the data is transformed these different ways. The kernel weighted fixed-effects estimates elasticities closer to the IV point estimates, but with much smaller standard errors.

Estimates of the same log-log model controlling for average log price of other producer products are numerically very similar, hence omitted. The kernel weighted estimates are similar to the own-price elasticity estimates from Table 1 in Hausman, Leonard and Zona (1994), which average around -2.5.

Figure 2 displays estimates from the factor model with products as rows, and the weighted-within estimator, with estimated confidence bands, allowing the estimated number of interactive fixed-effects terms to increase. The figure displays two things. First, there is evidence that up to approximately 9-10 factors may be appropriate. Second, that at each number of estimated factors, there is a significant shift in estimate when the weighted-within transformation is applied. Taking the model and theory proposed in this paper for granted, this implies a persistent debias with the weighted-within transformation.

Estimator	$\widehat{\beta}$ (St. err.)
Pooled OLS	1.304 (0.007)
Additive Fixed-effects	-0.069 (0.040)
Pooled IV	-2.148 (1.158)
Factor (Product (i) as rows)	-2.785 (0.055)
Factor (Store (j) as rows)	-0.098 (0.042)
Factor (Time (t) as rows)	-0.033 (0.003)
Weighted-within	-2.831 (0.054)

Table 4: Log-log demand elasticities (45 products, 48 stores, 110 fortnights).
Estimates for the standard errors are included in brackets. The factor model estimator uses five factors in all three cases.

7 Conclusion

This paper develops methods to generalise the interactive fixed-effect to multidimensional datasets with more than two dimensions. Theoretical results show that standard matrix methods can be applied to this setting but require additional knowledge of the data generating process. The multiplicative interactive fixed-effect error from the group fixed-effects and kernel methods show an improvement on the convergence rate of slope coefficient estimates and suggest a more robust approach to projecting fixed-effects. Simulations show finite sample properties when the sample size is allowed to grow and when it is fixed. These simulations exemplify how sensitive the two-dimensional methods are to model specification issues as simple as how to organise the dataset.

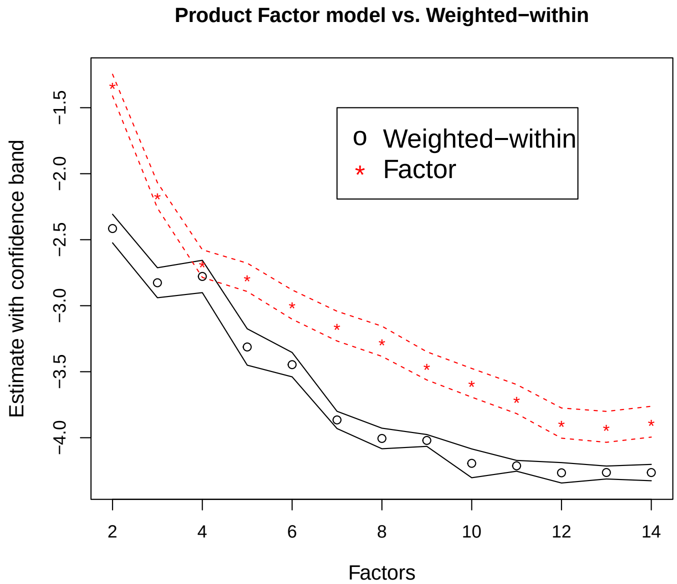

Figure 2: Elasticity estimates for increasing number of estimated factors

They also show the robustness of the group fixed-effects and kernel weighted fixed-effects estimators without having to make these same specifications. A method for inference with the weighted fixed-effects estimator is discussed, which can easily extend to the group fixed-effects estimator.

The model is applied to a log-log demand model for beer consumption. The application demonstrates that simply applying the two-dimensional factor model approach is sensitive to how dimensions are arranged to suit these estimators. The weighted-within transformation estimates elasticities close to the instrumental variable point estimates, but with substantially better precision.

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A Appendix: Proofs

A.1 Proofs of main paper results

Lemma A.1 (Regularity of proxy measures)**.** *Let* $\widehat{\varphi}_{i_n}^{(n)} \in \widehat{\Phi}_n$ *be the proxy space for the fixed-effects and let* $k(\cdot)$ *be a bounded kernel function from Assumption 5. If Assumption 6 holds, then as* $h_n \to 0$ *such that* $N_n h_n \to \infty$*,*

$$\operatorname*{plim}_{N_n \to \infty} \frac{1}{N_n h_n} \sum_{j=1}^{N_n} k\left(\frac{1}{h_n}\left\|\widehat{\varphi}_{i_n}^{(n)} - \widehat{\varphi}_{j_n}^{(n)}\right\|\right) = c_{i_n}^{(n)} \in (0, M] \tag{A.1}$$

for all $i_n \in 1, \ldots, N_n$*, where* $M < \infty$*.*

Proof of Lemma A.1. The supposition is that when

$$\operatorname*{plim}_{N_n \to \infty} \frac{M_n^h\left(\widehat{\varphi}_{i_n}^{(n^*)}\right)}{N_n h} = c_{i_n}^{(n)} \in (0, h^{-1}], \tag{A.2}$$

in Assumption 6 holds, then (A.1) holds. Take

$$\begin{aligned}\frac{1}{N_n h_n}\sum_{j=1}^{N_n} k\left(\frac{1}{h_n}\left\|\widehat{\varphi}_{i_n}^{(n)}-\widehat{\varphi}_{j_n}^{(n)}\right\|\right) &= \frac{1}{N_n h_n}\sum_{j_n:\widehat{\varphi}_{i_n}^{(n)}\in B_{h\varepsilon}(\widehat{\varphi}_{j_n}^{(n)})} k\left(\frac{1}{h_n}\left\|\widehat{\varphi}_{i_n}^{(n)}-\widehat{\varphi}_{j_n}^{(n)}\right\|\right)\\ &+ \frac{1}{N_n h_n}\sum_{j_n:\widehat{\varphi}_{i_n}^{(n)}\notin B_{h\varepsilon}(\widehat{\varphi}_{j_n}^{(n)})} k\left(\frac{1}{h_n}\left\|\widehat{\varphi}_{i_n}^{(n)}-\widehat{\varphi}_{j_n}^{(n)}\right\|\right)\\ &=: A_1 + A_2\end{aligned}$$

If the probability limit of A_1 or A_2 is strictly positive, then the sum is bounded below since both terms are weakly positive. The probability limit of both terms should be bounded above for the lemma to hold.

$$A_1 \geq \frac{1}{N_n h_n}\sum_{j_n} \mathbb{1}\{\widehat{\varphi}_{i_n}^{(n)} \in B_{h\varepsilon}(\widehat{\varphi}_{j_n}^{(n)})\}k\left(\varepsilon\right).$$

Set ε low enough such that $k\left(\varepsilon\right) > 0$. Then $\operatorname{plim} A_1 > 0$ by (A.2). $A_2 \geq 0$, hence no tighter lower bound needs to be shown for this term.

Now the upper bound.

$$\begin{aligned}A_1 &\leq \frac{1}{N_n h_n}\sum_{j_n} \mathbb{1}\{\widehat{\varphi}_{i_n}^{(n)} \in B_{h\varepsilon}(\widehat{\varphi}_{j_n}^{(n)})\} \max_j k\left(\frac{1}{h_n}\left\|\widehat{\varphi}_{i_n}^{(n)}-\widehat{\varphi}_{j_n}^{(n)}\right\|\right)\\ &\leq O(1)\frac{1}{N_n h_n}\sum_{j_n} \mathbb{1}\{\widehat{\varphi}_{i_n}^{(n)} \in B_{h\varepsilon}(\widehat{\varphi}_{j_n}^{(n)})\} = O_p(1),\end{aligned}$$

from bounded kernels. The probability limit of the last term is bounded by Assumption 6 as $O_p(1)$. For A_2, use the shorthand $\mathbb{1}(a,b)_{ij} = \mathbb{1}\left\{a < \left\|\widehat{\varphi}_{i_n}^{(n)}-\widehat{\varphi}_{j_n}^{(n)}\right\| < b\right\}$

$$\begin{aligned}&\frac{1}{N_n h_n}\sum_{j_n}\mathbb{1}(h_n\varepsilon,\infty)_{ij} k\left(\frac{1}{h_n}\left\|\widehat{\varphi}_{i_n}^{(n)}-\widehat{\varphi}_{j_n}^{(n)}\right\|\right)\\ &\quad=\frac{1}{N_n h_n}\sum_{m=1}^{\infty}\sum_{j_n}\mathbb{1}(mh_n\varepsilon,(m+1)h_n\varepsilon)_{ij} k\left(\frac{1}{h_n}\left\|\widehat{\varphi}_{i_n}^{(n)}-\widehat{\varphi}_{j_n}^{(n)}\right\|\right)\\ &\quad\leq\frac{1}{N_n h_n}\sum_{m=1}^{\infty}\sum_{j_n}\mathbb{1}(mh_n\varepsilon,(m+1)h_n\varepsilon)_{ij} k\left(m\varepsilon\right)\\ &\quad=\sum_{m=1}^{\infty} k\left(m\varepsilon\right)\frac{1}{N_n h_n}\sum_{j_n}\mathbb{1}(mh_n\varepsilon,(m+1)h_n\varepsilon)_{ij}\\ &\quad\leq\sum_{m=1}^{\infty} k\left(m\varepsilon\right)O_p(1)\\ &\quad\leq O_p(1)\int_0^{\infty} k\left(u\varepsilon\right)du\\ &\quad=O_p(1/\varepsilon)\int_0^{\infty} k^*\left(u\right)du = O_p(1/\varepsilon)\end{aligned}$$

The $k^*(u)$ comes from the equality for kernel functions $k^*(u) = \varepsilon k(\varepsilon u)$, where k^* is also a kernel function satisfying Assumption 5.(iii). Since $\varepsilon > 0$ and fixed, this term is also bounded above.

■

Lemma A.2 (Linear Combinations Bound)**.** *For the linear operator $\widetilde{H}_n \in \mathbb{R}^{r_n \times L}$, with bounded eigenvalues, define linear transformations of $\varphi^{(n)}_{i_n}$ as $\widetilde{\varphi}^{(n)}_{i_n} = \widetilde{H}_n \varphi^{(n)}_{i_n}$. When rank$(\widetilde{H}_n) = r_n < L$,*

$$\|\varphi^{(n)}_{i_n} - \varphi^{(n)}_{i'_n}\|^2 \leq \lambda_{\max}(\widetilde{H}_n) \left(\|\widetilde{\varphi}^{(n)}_{i_n} - \widehat{\widetilde{\varphi}}^{(n)}_{i_n}\|^2 + \|\widetilde{\varphi}^{(n)}_{i'_n} - \widehat{\widetilde{\varphi}}^{(n)}_{i'_n}\|^2 + \|\widehat{\widetilde{\varphi}}^{(n)}_{i_n} - \widehat{\widetilde{\varphi}}^{(n)}_{i'_n}\|^2 \right)$$

where $\lambda_{\max}$ is the maximum eigenvalue. When rank$(\widetilde{H}_n) = r_n = L$ then $\widetilde{H}_n$ is invertible and bounding $\|\varphi^{(n)}_{i_n} - \varphi^{(n)}_{i'_n}\|^2$ by $\|\widetilde{\varphi}^{(n)}_{i_n} - \widetilde{\varphi}^{(n)}_{i'_n}\|^2$ is straightforward.

Proof of Lemma A.2.

$$\begin{aligned}\|\varphi^{(n)}_{i_n} - \varphi^{(n)}_{i'_n}\|^2 \leq{}& \|\varphi^{(n)}_{i_n} - \widetilde{H}'_n \widetilde{\varphi}^{(n)}_{i_n}\|^2 + \|\varphi^{(n)}_{i'_n} - \widetilde{H}'_n \widetilde{\varphi}^{(n)}_{i'_n}\|^2 \\ &+ \|\widetilde{H}'_n \widetilde{\varphi}^{(n)}_{i_n} - \widetilde{H}'_n \widehat{\widetilde{\varphi}}^{(n)}_{i_n}\|^2 + \|\widetilde{H}'_n \widetilde{\varphi}^{(n)}_{i'_n} - \widetilde{H}'_n \widehat{\widetilde{\varphi}}^{(n)}_{i'_n}\|^2 \\ &+ \|\widetilde{H}'_n \widehat{\widetilde{\varphi}}^{(n)}_{i_n} - \widetilde{H}'_n \widehat{\widetilde{\varphi}}^{(n)}_{i'_n}\|^2\end{aligned}$$

The first term from the right hand side, $\|\varphi^{(n)}_{i_n} - \widetilde{H}'_n \widetilde{\varphi}^{(n)}_{i_n}\|^2 = \|(\mathbb{I}_L - \widetilde{H}'_n \widetilde{H}_n)(\varphi^{(n)}_{i_n} - \varphi^{(n)}_{i_n})\|^2 = 0$, from $\|(\mathbb{I}_L - \widetilde{H}'_n \widetilde{H}_n)(\varphi^{(n)}_{i_n} - \varphi^{(n)}_{i_n})\|^2 \leq \lambda_{\max}(\mathbb{I}_L - \widetilde{H}'_n \widetilde{H}_n)\|\varphi^{(n)}_{i_n} - \varphi^{(n)}_{i_n}\|^2 = 1 \cdot \|\varphi^{(n)}_{i_n} - \varphi^{(n)}_{i_n}\|^2$. Likewise, the second term is 0. Finally, use the matrix norm bound to bound the three remaining terms by the maximum eigenvalue of $\widetilde{H}_n$ factored out from each term. ■

The remaining proofs implicitly use Lemma A.2 and bound the space of each $\varphi^{(n)}_{i_n}$ without loss of generality with regard to the linear combination space that may be identified and estimated in the matrix method.

Proof of Proposition 2. In the following, let $\text{vec}_K(\widetilde{\mathbf{X}})$ be the $\prod_n N_n \times K$ matrix of vectorised covariates after the within-cluster transformations where each column is a vectorised transformed covariate. The vec operator on other variables is the standard vectorisation operator. Also let $N = \prod_n N_n$ and the subscript $i = 1, \ldots, N$ be the index for the vectorised data when i has no subscript. Then,

$$\begin{aligned}\beta_{KFE,\mathcal{C}} &= \left(\text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)} \text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}(\widetilde{\mathbf{Y}}) \\ &= \left(\text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)} \text{vec}_K(\widetilde{\mathbf{X}})' \left(\text{vec}_K(\widetilde{\mathbf{X}})\beta^0 + \text{vec}(\widetilde{\boldsymbol{\mathcal{A}}}) + \text{vec}(\widetilde{\boldsymbol{\varepsilon}})\right) \\ &= \beta^0 + \left(\text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)} \text{vec}_K(\widetilde{\mathbf{X}})' \left(\text{vec}(\widetilde{\boldsymbol{\mathcal{A}}}) + \text{vec}(\widetilde{\boldsymbol{\varepsilon}})\right),\end{aligned}$$

such that,

$$
\begin{aligned}
\left\|\beta_{KFE,\mathcal{C}}-\beta^0\right\| &= \left\| \left(\mathrm{vec}_K\big(\widetilde{\mathbf{X}}\big)'\mathrm{vec}_K\big(\widetilde{\mathbf{X}}\big)\right)^{(-1)} \mathrm{vec}_K\big(\widetilde{\mathbf{X}}\big)' \left(\mathrm{vec}\big(\widetilde{\boldsymbol{\mathcal{A}}}\big)+\mathrm{vec}\big(\widetilde{\varepsilon}\big)\right) \right\| \\
&\leq \|\kappa_N\| + \|\omega_N\|
\end{aligned}
$$

where

$$
\begin{aligned}
\|\kappa_N\| &:= \left\| \left(\mathrm{vec}_K\big(\widetilde{\mathbf{X}}\big)'\mathrm{vec}_K\big(\widetilde{\mathbf{X}}\big)\right)^{(-1)} \mathrm{vec}_K\big(\widetilde{\mathbf{X}}\big)'\mathrm{vec}\big(\widetilde{\boldsymbol{\mathcal{A}}}\big)\right\|; \\
\|\omega_N\| &:= \left\| \left(\mathrm{vec}_K\big(\widetilde{\mathbf{X}}\big)'\mathrm{vec}_K\big(\widetilde{\mathbf{X}}\big)\right)^{(-1)} \mathrm{vec}_K\big(\widetilde{\mathbf{X}}\big)'\mathrm{vec}\big(\widetilde{\varepsilon}\big)\right\|.
\end{aligned}
$$

$\|\omega_N\|$ is bounded at the parametric rate by standard argumetns, so $\|\kappa_N\|$ is the focus here.

Notice,

$$
\|\kappa_N\| \leq \left\| \left(\mathrm{vec}_K\big(\widetilde{\mathbf{X}}\big)'\mathrm{vec}_K\big(\widetilde{\mathbf{X}}\big)\right)^{(-1)} \right\| \left\|\mathrm{vec}_K\big(\widetilde{\mathbf{X}}\big)'\mathrm{vec}\big(\widetilde{\boldsymbol{\mathcal{A}}}\big)\right\|
$$

Focus on the right hand part, and let $\big\langle .,. \big\rangle_F$ be the Frobenius inner product,

$$
\left\|\mathrm{vec}_K\big(\widetilde{\mathbf{X}}\big)'\mathrm{vec}\big(\widetilde{\boldsymbol{\mathcal{A}}}\big)\right\| = \left\| \begin{bmatrix} \big\langle \widetilde{X}_1, \widetilde{\boldsymbol{\mathcal{A}}} \big\rangle_F \\ \vdots \\ \big\langle \widetilde{X}_K, \widetilde{\boldsymbol{\mathcal{A}}} \big\rangle_F \end{bmatrix} \right\| \leq \left\| \begin{bmatrix} \left|\sum_{i=1}^N \widetilde{X}_{i,1}\widetilde{\mathcal{A}}_i\right| \\ \vdots \\ \left|\sum_{i=1}^N \widetilde{X}_{i,K}\widetilde{\mathcal{A}}_i\right| \end{bmatrix} \right\| \tag{A.3}
$$

Let $\bar{\varphi}^{(n)}_{i_n^*,\ell}$ denote now the estimate of $\varphi^{(n)}_{i_n,\ell}$ according the the kernel weighted estimator. The entries for each k in (A.3) can be written as,

$$
\begin{aligned}
\sum_{i=1}^{N} \widetilde{X}_{i,k}\widetilde{\mathcal{A}}_i &= \frac{1}{N}\sum_{i_1,\dots,i_d} X_{i_1,\dots,i_d;k}\sum_{\ell=1}^{L}\prod_n \left(\varphi^{(n)}_{i_n,\ell} - \bar{\varphi}^{(n)}_{i_n^*,\ell}\right) \\
&= \frac{1}{N}\sum_{i_1,\dots,i_d} X_{i_1,\dots,i_d;k}\sum_{\ell=1}^{L}\prod_n \sum_{j_n} W^{(n)}_{i_n,j_n}\left(\varphi^{(n)}_{i_n,\ell} - \varphi^{(n)}_{j_n,\ell}\right) \\
&= \frac{1}{N}\sum_{i_1,\dots,i_d}\sum_{j_1,\dots,j_d}\sum_{\ell=1}^{L}\prod_n \sqrt{W^{(n)}_{i_n,j_n}} X_{i_1,\dots,i_d;k}\prod_n \sqrt{W^{(n)}_{i_n,j_n}}\left(\varphi^{(n)}_{i_n,\ell} - \varphi^{(n)}_{j_n,\ell}\right) \\
&\leq \frac{1}{N}\left(\sum_{i_1,\dots,i_d}\sum_{j_1,\dots,j_d}\sum_{\ell=1}^{L}\prod_n W^{(n)}_{i_n,j_n} X^2_{i_1,\dots,i_d;k}\right)^{1/2} \times \dots \\
&\dots \times \left(\sum_{i_1,\dots,i_d}\sum_{j_1,\dots,j_d}\sum_{\ell=1}^{L}\prod_n W^{(n)}_{i_n,j_n}\left(\varphi^{(n)}_{i_n,\ell} - \varphi^{(n)}_{j_n,\ell}\right)^2\right)^{1/2} \\
&\leq \left(\frac{1}{N}\sum_{\ell=1}^{L}\sum_{i_1,\dots,i_d} X^2_{i_1,\dots,i_d;k}\sum_{j_1,\dots,j_d}\prod_n W^{(n)}_{i_n,j_n}\right)^{1/2} \times \dots \\
&\dots \times \left(\frac{1}{N}\sum_{i_1,\dots,i_d}\sum_{j_1,\dots,j_d}\prod_n W^{(n)}_{i_n,j_n}\left\|\varphi^{(n)}_{i_n} - \varphi^{(n)}_{j_n}\right\|^2\right)^{1/2} \\
&= \sqrt{L}O_p(1)\left(\frac{1}{N}\sum_{i_1,\dots,i_d}\sum_{j_1,\dots,j_d}\prod_n W^{(n)}_{i_n,j_n}\left\|\varphi^{(n)}_{i_n} - \varphi^{(n)}_{j_n}\right\|^2\right)^{1/2}
\end{aligned}
$$

The last line comes from bounded second moments in X and from weights summing to 1. The final term can be treated separately for each n.

$$
\frac{1}{N_n}\sum_{i_n}\sum_{j_n} W^{(n)}_{i_n,j_n}\left\|\varphi^{(n)}_{i_n} - \varphi^{(n)}_{j_n}\right\|^2
$$

Use the triangle inequality,

$$
\left\|\varphi^{(n)}_{i_n} - \varphi^{(n)}_{j_n}\right\| \leq \left\|\varphi^{(n)}_{i_n} - \widehat{\varphi}^{(n)}_{i_n}\right\| + \left\|\varphi^{(n)}_{j_n} - \widehat{\varphi}^{(n)}_{j_n}\right\| + \left\|\widehat{\varphi}^{(n)}_{i_n} - \widehat{\varphi}^{(n)}_{j_n}\right\|
$$

to bound this as

$$
\frac{1}{N_n}\sum_{i_n}\sum_{j_n} W^{(n)}_{i_n,j_n} O\left(\left\|\varphi^{(n)}_{i_n} - \widehat{\varphi}^{(n)}_{i_n}\right\|^2 + \left\|\varphi^{(n)}_{j_n} - \widehat{\varphi}^{(n)}_{j_n}\right\|^2 + \left\|\widehat{\varphi}^{(n)}_{i_n} - \widehat{\varphi}^{(n)}_{j_n}\right\|^2\right),
$$

with a few applications of the Cauchy-Schwarz inequality.

Take the first of these terms,

$$\frac{1}{N_n}\sum_{i_n}\sum_{j_n} W^{(n)}_{i_n,j_n}\left\|\varphi^{(n)}_{i_n}-\widehat{\varphi}^{(n)}_{i_n}\right\|^2 = \frac{1}{N_n}\sum_{i_n}\left\|\varphi^{(n)}_{i_n}-\widehat{\varphi}^{(n)}_{i_n}\right\|^2\sum_{j_n} W^{(n)}_{i_n,j_n}$$
$$= O_p(C_n^{-2})$$

Likewise, the second term is also $O_p(C_n^{-2})$.

The third term, with the shorthand $K^{(n)}_{ij,h} = k\left(\frac{1}{h_n}\left\|\widehat{\varphi}^{(n)}_{i_n}-\widehat{\varphi}^{(n)}_{j_n}\right\|\right)$, and $\mathbb{1}(a,b)_{ij} = \mathbb{1}\Big\{a < \|\widehat{\varphi}^{(n)}_{i_n}-\widehat{\varphi}^{(n)}_{j_n}\| < b\Big\}$,

$$\begin{aligned}
\frac{1}{N_n}\sum_{i_n}\sum_{j_n} W^{(n)}_{i_n,j_n}\left\|\widehat{\varphi}^{(n)}_{i_n}-\widehat{\varphi}^{(n)}_{j_n}\right\|^2 &= \frac{1}{N_n^2 h_n}\sum_{i_n}\sum_{j_n} K^{(n)}_{ij,h}\left\|\widehat{\varphi}^{(n)}_{i_n}-\widehat{\varphi}^{(n)}_{j_n}\right\|^2 \qquad \text{(A.4)}\\
&= \frac{1}{N_n^2 h_n}\sum_{i_n}\sum_{j_n}\mathbb{1}(0,h_n)_{ij}K^{(n)}_{ij,h}\left\|\widehat{\varphi}^{(n)}_{i_n}-\widehat{\varphi}^{(n)}_{j_n}\right\|^2\\
&+ \frac{1}{N_n^2 h_n}\sum_{i_n}\sum_{j_n}\sum_{m=1}^{\infty}\mathbb{1}(mh_n,(m+1)h_n)_{ij}K^{(n)}_{ij,h}\left\|\widehat{\varphi}^{(n)}_{i_n}-\widehat{\varphi}^{(n)}_{j_n}\right\|^2\\
&\le \frac{1}{N_n^2 h_n}\sum_{i_n}\sum_{j_n}\mathbb{1}(0,h_n)_{ij}K^{(n)}_{ij,h}h_n^2\\
&+ \frac{1}{N_n^2 h_n}\sum_{i_n}\sum_{j_n}\sum_{m=1}^{\infty}\mathbb{1}(mh_n,(m+1)h_n)_{ij}k(m)(m+1)^2h_n^2\\
&= O(h_n^2)+O(h_n^2)\sum_{m=1}^{\infty}(m+1)^2k(m)\\
&\le O(h_n^2)+O(h_n^2)\int_0^{\infty}(m^2+2m+1)k(m) = O(h_n^2)
\end{aligned}$$

The last line comes from bounded first and second moments of the normed proxy measure space, and integrability of the kernel function. The bound $\sum_{i_n}\sum_{j_n}\mathbb{1}(mh_n,(m+1)h_n)_{ij} \lesssim N_n^2 h_n$ is a direct corollary of Assumption 6, noting only the upper bound is required here.

Hence, for each n

$$\frac{1}{N_n}\sum_{i_n}\sum_{j_n} W^{(n)}_{i_n,j_n}\left\|\varphi^{(n)}_{i_n}-\varphi^{(n)}_{j_n}\right\|^2 \le \left(O_p\left(C_n^{-2}\right)+O_p\left(h_n^2\right)\right).$$

Taking the product of these terms over all dimensions then forms the statement of the result.

■

Assumption A.1 (Mean regularity of proxy measures)**.** *Let $\widehat{\varphi}^{(n)}_{i_n} \in \widehat{\Phi}_n$ and redefine $M_n^{\varepsilon}\left(\widehat{\varphi}^{(n)}_{i_n}\right)$ as*

$$M_n^h\left(\widehat{\varphi}^{(n)}_{i_n}\right) := \sum_{j_n=1}^{N_n}\mathbb{1}\left(\widehat{\varphi}^{(n)}_{j_n}\in B_{he}\left(\widehat{\varphi}^{(n)}_{i_n}\right)\right),$$

where $B_{he}(x)$ is the he-neighbourhood around x. Then for any $e > 0$, as $h \to 0$ such that $N_n h \to \infty$,

$$\operatorname*{plim}_{N_n \to \infty} \frac{1}{N_n} \sum_{i_n=1}^{N_n} \frac{M_n^h \left(\widehat{\varphi}_{i_n}^{(n^*)} \right)}{N_n h} = c^{(n)} \in (0, h^{-1}].$$

Lemma A.3 (KFE consistency under mean bounds)**.** *Take proposition 2, with Assumption 6 replaced by Assumption A.1. In addition, assume the proxy measures sampled from $\widehat{\Phi}_n$ have bounded third moments. Then,*

$$\left\| \widehat{\beta}_{KER,\mathcal{W}} - \beta^0 \right\| = \sqrt{L} O_p \left(\prod_{n^* \in \mathcal{M}'} O_p\left(C_{n^*}^{-1}\right) + O_p\left(h_{n^*}^{5/4} N_{n^*}^{1/2}\right) \right) + O_p \left(\prod_{n=1}^{d} \frac{1}{\sqrt{N_n}} \right).$$

For $h_n \lesssim O\left(C_n^{-1} N_n^{-1/2}\right)^{4/5}$, when $C_n^{-1} N_n^{-1/2} \to 0$, this reduces to

$$\left\| \widehat{\beta}_{KER,\mathcal{W}} - \beta^0 \right\| = \sqrt{L} O_p \left(\prod_{n^* \in \mathcal{M}'} O_p\left(C_{n^*}^{-1}\right) \right) + O_p \left(\prod_{n=1}^{d} \frac{1}{\sqrt{N_n}} \right).$$

Proof of Lemma A.3. The condition in Assumption A.1 impacts how (A.4) can be bound. The other terms in the proof of Proposition 2 are unchanged.

$$\begin{aligned}
\frac{1}{N_n} \sum_{i_n} \sum_{j_n} W_{i_n, j_n}^{(n)} \left\| \widehat{\varphi}_{i_n}^{(n)} - \widehat{\varphi}_{j_n}^{(n)} \right\|^2 &= \frac{1}{N_n} \sum_{i_n} \sum_{j_n} \frac{K_{ij,h}^{(n)}}{\sum_{j_n} K_{ij,h}^{(n)}} \left\| \widehat{\varphi}_{i_n}^{(n)} - \widehat{\varphi}_{j_n}^{(n)} \right\|^2 \\
&\leq \left(\frac{1}{N_n} \sum_{i_n} \sum_{j_n} K_{ij,h}^{(n)\,2} \left\| \widehat{\varphi}_{i_n}^{(n)} - \widehat{\varphi}_{j_n}^{(n)} \right\|^4 \right)^{1/2} \left(\frac{1}{N_n} \sum_{i_n} \sum_{j_n} \frac{1}{\left(\sum_{j_n} K_{ij,h}^{(n)} \right)^2} \right)^{1/2}
\end{aligned}$$

The first term can be bounded,

$$\begin{aligned}
\frac{1}{N_n} \sum_{i_n} \sum_{j_n} K_{ij,h}^{(n)\,2} \left\| \widehat{\varphi}_{i_n}^{(n)} - \widehat{\varphi}_{j_n}^{(n)} \right\|^4 &= \frac{O(h_n^2)}{N_n h_n} \sum_{i_n} \sum_{j_n} K_{ij,h}^{(n)} \left\| \widehat{\varphi}_{i_n}^{(n)} - \widehat{\varphi}_{j_n}^{(n)} \right\|^3 \\
&= \frac{O(h_n^2)}{N_n h_n} \sum_{i_n} \sum_{j_n} \mathbb{1}(0, h_n)_{ij} K_{ij,h}^{(n)} \left\| \widehat{\varphi}_{i_n}^{(n)} - \widehat{\varphi}_{j_n}^{(n)} \right\|^3 \\
&+ \frac{O(h_n^2)}{N_n h_n} \sum_{i_n} \sum_{j_n} \sum_{m=1}^{\infty} \mathbb{1}(m h_n, (m+1) h_n)_{ij} K_{ij,h}^{(n)} \left\| \widehat{\varphi}_{i_n}^{(n)} - \widehat{\varphi}_{j_n}^{(n)} \right\|^3 \\
&\leq O_p(h_n^5 N_n) + \frac{O(h_n^5)}{N_n h_n} \sum_{i_n} \sum_{j_n} \sum_{m=1}^{\infty} \mathbb{1}(m h_n, (m+1) h_n)_{ij} k(m)(m+1)^3 \\
&\leq O_p(h_n^5 N_n) + O_p(h_n^5 N_n) \int_0^{\infty} (u^3 + 3u^2 + 3u + 1) k(u) du = O_p(h_n^5 N_n).
\end{aligned}$$

The last line comes from the additional requirement that the proxy measures have bounded third moments. This bounds the first term as $O_p(h_n^{5/2} N_n^{1/2})$. The second term can be simply bound by $O(N_n^{1/2})$. Apply the square root to leave the final bound, $O_p(h_n^{5/4} N_n^{1/2})$.

■

A.2 Iterative Estimator

Assumption 6 can be significantly relaxed with the use of an iterative projection of fixed-effects. The following estimation procedure allows for less restrictive sampling of the proxy measures used to form projection weights. Namely, the sampling restriction can be entry-wise, i.e. on scalars, rather than on the neighbourhoods of entire vectors. The procedure is general, and can use many weights, or even clusters, in place of the kernel weights used to describe the estimator below.

Every tensor has a higher-order singular value decomposition (HOSVD),

$$\mathcal{A} = \mathcal{S} \times (U^{(1)}, \ldots, U^{(d)})$$

where $\mathcal{S} \in \mathbb{R}^{\times_{n=1}^{d} r_n}$ and $U^{(n)} \in \mathbb{R}^{N_n \times r_n}$, where r_n is the n-th compononent of the multilinear rank. When written in the above form where $U^{(n)} \in \mathbb{R}^{N_n \times r_n}$ rather than $U^{(n)} \in \mathbb{R}^{N_n \times N_n}$, this can be referred to as the compact HOSVD. The iterative estimator works as follows.

1. Obtain proxies for $U^{(n)}$, called $\widehat{U}^{(n)}$ for each $n = 1, \ldots, d$. Use these to form weights, for $m_n = 1, \ldots, r_n$;

$$W_{nm_n,i_nj_n} := \frac{k\left(\frac{1}{h_n}\left(\widehat{U}^{(n)}_{i_n,m_n} - \widehat{U}^{(n)}_{j_n,m_n}\right)^2\right)}{\sum_{i'_n=1}^{N_n} k\left(\frac{1}{h_n}\left(\widehat{U}^{(n)}_{i_n,m_n} - \widehat{U}^{(n)}_{i'_n,m_n}\right)^2\right)} \qquad \text{for each } n = 1, \ldots, d. \tag{A.5}$$

2. For each $m_n = 1, \ldots, r_n$ project out weighted means from $\boldsymbol{Y}$ and $\boldsymbol{X}_k$ for each $k = 1, \ldots, K$ along each dimension, $n = 1, \ldots, d$ as follows,

$$\widetilde{\mathcal{A}} = \mathcal{A} \times \left(\prod_{m_1=1}^{r_1} (\mathbb{I} - W_{1m_1}), \ldots, \prod_{m_d=1}^{r_d} (\mathbb{I} - W_{dm_d})\right) \tag{A.6}$$

3. Perform pooled OLS of $\widetilde{\boldsymbol{Y}}$ on $\widetilde{\boldsymbol{X}}$. Call the estimator $\widehat{\beta}_{IK}$, for iterative kernel.

From the representation in (A.6) the proof techniques from previous results follow straightforwardly. As is proposed in the main text, proxies $\widehat{U}^{(n)}$ can be estimated using the matrix method with each dimension used as rows. Then results from Proposition A.1 in Bai (2009) can be used to bound the estimation error of these proxies with respect to the true $U^{(n)}$. Greater care is needed for the fact Proposition A.1 in Bai (2009) is an up-to-rotations result that is not required in previous results, hence this is taken care of specifically in the proof.

Assumption A.2 (Regularity of proxy measures)**.** *Define* $M_n^h\left(\widehat{U}^{(n)}_{i_n,m_n}\right)$ *as*

$$M_n^h\left(\widehat{U}^{(n)}_{i_n,m_n}\right) := \sum_{j_n=1}^{N_n} \mathbb{1}\left(\widehat{U}^{(n)}_{j_n,m_n} \in B_{he}\left(\widehat{U}^{(n)}_{i_n,m_n}\right)\right),$$

where $B_{he}(x)$ is an he-neighbourhood around x. Then for each dimension n, for any $e > 0$, as $h \to 0$ such that $N_n h \to \infty$,

$$\operatorname*{plim}_{N_n \to \infty} \frac{M_n^h\left(\widehat{U}_{i_n,m_n}^{(n)}\right)}{N_n h} = c_{i_n}^{(n)} \in (0, h^{-1}]$$

for all $\widehat{U}_{i_n,m_n}^{(n)}$. Further, let $\widehat{U}_{i_n,m_n}^{(n)}$ have a bounded second moment.

Proposition A.1 (Upper bound on iterative estimator)**.** *Let Assumption 5, Assumption 7 specifically for the iterative transformation, and Assumption A.2 hold. For the higher order singular value decomposition of the fixed-effects term, let $\frac{1}{N_{n^*}} \sum_{i_{n^*}} \left\| U_{i_{n^*}}^{(n^*)} - H^{(n^*)} \widehat{U}_{i_{n^*}}^{(n^*)} \right\|^2 = O_p(C_{n^*}^{-2})$ for $n^* \in \mathcal{M}'$ with $C_{n^*}^{-2} \to 0$, and $\frac{1}{N_{n'}} \sum_{i_{n'}} \left\| U_{i_{n'}}^{(n')} - H^{(n')} \widehat{U}_{i_{n'}}^{(n')} \right\|^2 = O_p(1)$ for $n' \notin \mathcal{M}'$, where $\mathcal{M}'$ is a non-empty subset of dimensions. $H^{(n)}$ is an invertible $r_n \times r_n$ matrix. Let h_n be the bandwidth parameter from Assumption A.2. Then,*

$$\left\| \widehat{\beta}_{IK,\mathcal{W}} - \beta^0 \right\| = O_p\Big(\prod_n r_n^{1/2} (C_n^{-1} + r_n^{1/2} h_n) \Big) + O_p\left(\prod_{n=1}^d \frac{1}{\sqrt{N_n}} \right).$$

For $h_n \lesssim O(C_n^{-1} r_n^{-1/2})$ this reduces to

$$\left\| \widehat{\beta}_{IK,\mathcal{W}} - \beta^0 \right\| = \prod_n r_n^{1/2} O_p\left(\prod_{n^* \in \mathcal{M}'} O_p\left(C_{n^*}^{-1}\right) \right) + O_p\left(\prod_{n=1}^d \frac{1}{\sqrt{N_n}} \right).$$

Proof of Proposition A.1.

$$\frac{1}{N} \sum_{i_1,\dots,i_d} \widetilde{X}_{i_1,\dots,i_d} \sum_{m_1}^{r_1} \cdots \sum_{m_d}^{r_d} \mathcal{S}_{m_1,\dots,m_d} \prod_{m_1'=1}^{r_1} (U_{i_1,m_1}^{(1)} - W_{1m_1',i_1}' U_{\cdot,m_1}^{(1)}) \cdots \prod_{m_d'=1}^{r_d} (U_{i_d,m_d}^{(d)} - W_{dm_d',i_d}' U_{\cdot,m_d}^{(d)})$$

$$= \frac{1}{N} \sum_{i_1,\dots,i_d} \sum_{j_1\dots,j_d} \widetilde{X}_{i_1,\dots,i_d} \sum_{m_1}^{r_1} \cdots \sum_{m_d}^{r_d} \mathcal{S}_{m_1,\dots,m_d} \prod_{m_1'=1}^{r_1} W_{1m_1',i_1j_1} (U_{i_1,m_1}^{(1)} - U_{j_1,m_1}^{(1)}) \dots$$

$$= \frac{1}{N} \sum_{i_1,\dots,i_d} \sum_{j_1\dots,j_d} \widetilde{X}_{i_1,\dots,i_d} \prod_{m_1'=1}^{r_1} \sqrt{W_{1m_1',i_1j_1}} \cdots \sum_{m_1}^{r_1} \cdots \sum_{m_d}^{r_d} \mathcal{S}_{m_1,\dots,m_d} \prod_{m_1'=1}^{r_1} \sqrt{W_{1m_1',i_1j_1}} (U_{i_1,m_1}^{(1)} - U_{j_1,m_1}^{(1)}) \dots$$

By Cauchy-Schwarz, this can be bound by

$$\left(\frac{1}{N} \sum_{i_1,\dots,i_d} \sum_{m_1}^{r_1} \cdots \sum_{m_d}^{r_d} \widetilde{X}_{i_1,\dots,i_d}^2 \sum_{j_1\dots,j_d} \prod_{m_1'=1}^{r_1} W_{1m_1',i_1j_1} \cdots \prod_{m_d'=1}^{r_d} W_{dm_d',i_dj_d} \right)^{1/2} \dots \tag{A.7}$$

$$\dots \left(\frac{1}{N} \sum_{i_1,\dots,i_d} \sum_{j_1\dots,j_d} \sum_{m_1}^{r_1} \cdots \sum_{m_d}^{r_d} \mathcal{S}_{m_1,\dots,m_d}^2 \prod_{m_1'=1}^{r_1} W_{1m_1',i_1j_1} (U_{i_1,m_1}^{(1)} - U_{j_1,m_1}^{(1)})^2 \dots \right)^{1/2}$$

Focus on the first term in (A.7) and note

$$\sum_{j_1 \dots, j_d} \prod_{m_1'=1}^{r_1} W_{1m_1', i_1 j_1} \cdots \prod_{m_d'=1}^{r_d} W_{dm_d', i_d j_d} = \sum_{j_1} \prod_{m_1'=1}^{r_1} W_{1m_1', i_1 j_1} \cdots \sum_{j_d} \prod_{m_d'=1}^{r_d} W_{dm_d', i_d j_d}$$

such that for any n,

$$\begin{aligned}
\sum_{j_n} \prod_{m_n'=1}^{r_1} W_{nm_n', i_n j_n} &\leq \left(\sum_{j_n} W_{n1, i_n j_n}^2 \right)^{1/2} \left(\sum_{j_n} \prod_{m_n' \neq 1} W_{nm_n', i_n j_n}^2 \right)^{1/2} \\
\text{(weights in [0,1])} \quad &\leq \left(\sum_{j_n} W_{n1, i_n j_n} \right)^{1/2} \left(\sum_{j_n} \prod_{m_n' \neq 1} W_{nm_n', i_n j_n} \right)^{1/2} \\
\text{(weights sum to 1)} \quad &\leq 1 \cdot \left(\sum_{j_n} W_{n2, i_n j_n}^2 \right)^{1/2} \left(\sum_{j_n} \prod_{m_n' \neq \{1,2\}} W_{nm_n', i_n j_n}^2 \right)^{1/2} \cdots
\end{aligned}$$

which is subsequently bound by 1. By bounded second moments of X post transformation, this bounds the first term as $O_p\left(\prod_n r_n^{1/2}\right)$.

Now for the second term in (A.7). For singular values, $\mathcal{S}_{m_1,\dots,m_d}$, bounded in probability this term is bounded as,

$$\begin{aligned}
&\max_{m_1,\dots,m_d} \mathcal{S}_{m_1,\dots,m_d}^2 \prod_{n=1}^{d} \sum_{m_n}^{r_n} \frac{1}{N_n} \sum_{i_n} \sum_{j_n} \prod_{m_n'=1}^{r_n} W_{nm_n', i_n j_n} \big(U_{i_n, m_n}^{(n)} - U_{j_n, m_n}^{(n)}\big)^2 \\
&\qquad = O_p(1) \prod_{n=1}^{d} \sum_{m_n}^{r_n} \frac{1}{N_n} \sum_{i_n} \sum_{j_n} \prod_{m_n'=1}^{r_n} W_{nm_n', i_n j_n} \big(U_{i_n, m_n}^{(n)} - U_{j_n, m_n}^{(n)}\big)^2
\end{aligned}$$

Focus for each n, where the superscript notation in $H^{(n)}$ is suppressed for brevity,

$$\begin{aligned}
&\sum_{m_n}^{r_n} \frac{1}{N_n} \sum_{i_n, j_n} \prod_{m_n'=1}^{r_n} W_{nm_n', i_n j_n} \big(U_{i_n, m_n}^{(n)} - U_{j_n, m_n}^{(n)}\big)^2 \leq \dots \\
&\qquad \dots \leq \sum_{m_n}^{r_n} \frac{3}{N_n} \sum_{i_n, j_n} \prod_{m_n'=1}^{r_n} W_{nm_n', i_n j_n} \big(U_{i_n, m_n}^{(n)} - H_{m_n}' \widehat{U}_{i_n}^{(n)}\big)^2 \\
&\qquad\quad + \sum_{m_n}^{r_n} \frac{3}{N_n} \sum_{i_n, j_n} \prod_{m_n'=1}^{r_n} W_{nm_n', i_n j_n} \big(U_{j_n, m_n}^{(n)} - H_{m_n}' \widehat{U}_{j_n}^{(n)}\big)^2 \\
&\qquad\quad + \sum_{m_n}^{r_n} \frac{3}{N_n} \sum_{i_n, j_n} \prod_{m_n'=1}^{r_n} W_{nm_n', i_n j_n} \big(H_{m_n}' \widehat{U}_{i_n}^{(n)} - H_{m_n}' \widehat{U}_{j_n}^{(n)}\big)^2
\end{aligned}$$

by Jensen's inequality. The first term,

$$\begin{aligned}
\frac{3}{N_n} \sum_{i_n} \sum_{m_n}^{r_n} \big(U_{i_n, m_n}^{(n)} - H_{m_n}' \widehat{U}_{i_n}^{(n)}\big)^2 \sum_{j_n} \prod_{m_n'=1}^{r_n} W_{nm_n', i_n j_n} &\leq \frac{3}{N_n} \sum_{i_n} \big\| U_{i_n}^{(n)} - H' \widehat{U}_{i_n}^{(n)} \big\|^2 \cdot 1 \\
&= O_p(C_n^{-2})
\end{aligned}$$

Likewise, the second term is $O_p(C_n^{-2})$.

The final term,

$$\frac{3}{N_n}\sum_{i_n,j_n}\prod_{m'_n=1}^{r_n} W_{nm'_n,i_nj_n}\sum_{m_n}^{r_n}\Big(H'_{m_n}(\widehat{U}^{(n)}_{i_n}-\widehat{U}^{(n)}_{j_n})\Big)^2 \leq \dots$$

$$\dots \leq \frac{3}{N_n}\sum_{i_n,j_n}\prod_{m'_n=1}^{r_n} W_{nm'_n,i_nj_n}\|\widehat{U}^{(n)}_{i_n}-\widehat{U}^{(n)}_{j_n}\|^2\sum_{m_n}^{r_n}\|H_{m_n}\|^2$$

$$= O_p(r_n)\frac{1}{N_n}\sum_{i_n,j_n}\sum_{m_n}^{r_n}\prod_{m'_n=1}^{r_n} W_{nm'_n,i_nj_n}(\widehat{U}^{(n)}_{i_nm_n}-\widehat{U}^{(n)}_{j_nm_n})^2,$$

where the last bound results from H being invertible, hence has bounded singular values. Since $\sum_{m_n}^{r_n}\|H_{m_n}\|^2 = \sum_{m_n}^{r_n}\sum_{m'_n}^{r_n} H^2_{m'_n} = \sum_{r=1}^{r_n}\sigma_r^2(H)$, where $\sigma_r^2(H)$ are the bounded singular values of H (bounded from invertibility of H). Without loss, assume the ordering of m_n and m'_n from the sum and product in this last expression are the same. Then, for each m_n

$$\frac{1}{N_n}\sum_{i_n,j_n}\prod_{m'_n=1}^{r_n} W_{nm'_n,i_nj_n}(\widehat{U}^{(n)}_{i_nm_n}-\widehat{U}^{(n)}_{j_nm_n})^2 = \frac{1}{N_n}\sum_{i_n,j_n} W_{nm_n,i_nj_n}(\widehat{U}^{(n)}_{i_nm_n}-\widehat{U}^{(n)}_{j_nm_n})^2\prod_{m'_n\neq m_n} W_{nm'_n,i_nj_n}$$

$$\leq \frac{1}{N_n}\sum_{i_n,j_n} W_{nm_n,i_nj_n}(\widehat{U}^{(n)}_{i_nm_n}-\widehat{U}^{(n)}_{j_nm_n})^2\cdot 1$$

As in the proof of Proposition 2, this final term is bound as $O_p(h_n^2)$, hence, the second term of (A.7) is $O_p(r_nh_n^2)$.

Thus, consistency is at the rate $O_p\Big(\prod_n r_n^{1/2}(C_n^{-1}+r_n^{1/2}h_n)\Big)$.

■

B Appendix: Additional Results

B.1 Group fixed-effects

This section describes the group fixed-effects estimator. Take again the model in array notation,

$$\mathbf{Y} = \sum_{k=1}^{K}\mathbf{X}_k\beta_k^0 + \boldsymbol{\mathcal{A}} + \boldsymbol{\varepsilon}.$$

A cluster assignment, $\mathcal{C}$, is a length d list of partition matrices, $C_n \in \{0,1\}^{N_n\times G_n}$, where G_n is the number of clusters in dimension n and each entry of C_n is a binary indicator of a unit's membership to a given cluster. Clusters are assigned separately along each dimension. Let $\Theta_{\mathcal{C}}$ be the space of group fixed-effects parameters associated to cluster assignment $\mathcal{C}$. Each $\boldsymbol{\theta}\in\Theta_{\mathcal{C}}$ is an ordered set of size d of $\bigtimes_{n=1}^{d} N_n$ tensors. For each n in $\{1,\dots,d\}$, the tensor θ_n varies freely over dimensions $\{1,\dots,n-1\}$ and $\{n+1,\dots,d\}$ but is constant within each cluster along dimension n.[9] The objective function for the group fixed-effect estimation of β under cluster assignment $\mathcal{C}$ is

$$Q(\beta, \mathcal{C}) = \min_{\theta \in \Theta_{\mathcal{C}}} \left\| \mathbf{Y} - \sum_{k=1}^{K} \mathbf{X}_k \beta_k - \sum_{n=1}^{d} \theta_n \right\|_F^2 \tag{B.1}$$

and $\widehat{\beta}_{GFE,\mathcal{C}} := \operatorname{argmin}_{\beta \in \mathbb{R}^K} Q(\beta, \mathcal{C})$.

The minimum within (B.1) is obtained from the within-cluster transformation in (3) from the Introduction. Remark B.1 below details this objective function for the three-dimensional setting for clarity. It should be clear that the parameter space $\Theta_{\mathcal{C}}$ is indexed by cluster assignment $\mathcal{C}$ because this assignment defines how the parameters may vary. That is, this is the estimated parameter space under a specific group fixed-effects estimator, which may only be an approximation of the true parameter space.

The within-cluster transformation can be stated in more general fashion as follows. Take M_n to be an $N_n \times N_n$ matrix defined as $\mathbb{I}_{N_n} - C_n(C_n'C_n)^{\dagger}C_n'$, where † is the Moore-Penrose generalised inverse. Then, the within-cluster transformation can be formed by the following series of n-mode products, $\mathbf{Y} \times_1 M_1 \times_2 M_2 \times_3 \cdots \times_d M_d$. This sequentially differences out the group specific means from each dimension separately. Then the Frisch-Waugh-Lovell theorem straightforwardly applies.

Cluster assignments may be known or estimated. In the case these are not known, use the estimates $\widehat{\varphi}^{(n)}$ from estimating (8) for each $n = 1, \ldots, d$ as proxies for clustering. That is, repeat the matrix method estimation for each flattening $n = 1, \ldots, d$, and use the fixed-effects estimates in, for example, the K-means algorithm to form clusters.

Assumption B.1 (Clustering)**.**

Let $j_n(i_n)$ be any unit in the same cluster as i_n from cluster assignment $\mathcal{C}$. Then,

(i). For all n as $N_n \to \infty$, $\frac{1}{N_n}\sum_{i_n=1}^{N_n} \left\| \varphi_{i_n}^{(n)} \right\|^2 \lesssim O_p(1)$, and,

(ii). For a non-empty subset $\mathcal{M} \subset \{1, \ldots, d\}$ take for any $n^* \in \mathcal{M}$ a sequence $\xi_{N_{n^*}} \to 0$ as $N_{n^*} \to \infty$. Then,

$$\frac{1}{N_{n^*}} \sum_{i_n=1}^{N_{n^*}} \left\| \varphi_{i_{n^*}}^{(n^*)} - \varphi_{j_{n^*}(i_{n^*})}^{(n^*)} \right\|^2 = O_p(\xi_{N_{n^*}})$$

Assumption B.1.(i) restricts fixed-effects parameter space to have finite second moments. This implies that as $\{N_1, \ldots, N_d\} \to \infty$, cluster allocations cannot increase too disparately in the underlying parameter space. Assumption B.1.(ii) states that for at least one dimension the

[9] This parameter space is exemplified in Remark B.1 for the three dimensional setting for clarity.

clustering procedure finds matches with asymptotically negligible difference in the underlying parameter space. Since cluster assignments are not always estimated, and can be given extraneously, it is useful to state Assumption B.1 in generic terms that ignore these clustering mechanics.

Below is a refinement to the regularity conditions from Section 3.1 to cater for the within-cluster transformation.

Assumption B.2 (Regularity conditions). *Let $\widetilde{T}_{i_1,\ldots,i_d}$ be the entries of tensor $\mathbf{T}$ after the group fixed-effects from the minimiser of* (B.1) *are differenced out. Then,*

(i). $\left(\frac{1}{\prod_n N_n}\sum_{i_1}\cdots\sum_{i_d}\widetilde{X}_{i_1,\ldots,i_d}\widetilde{X}'_{i_1,\ldots,i_d}\right) = O_p(1)$ *converges to a nonrandom positive definite matrix as* $N_1,\ldots,N_d \to \infty$.

(ii). $\frac{1}{\prod_n N_n}\sum_{i_1}\cdots\sum_{i_d}\widetilde{X}_{i_1,\ldots,i_d}\varepsilon_{i_1,\ldots,i_d} = O_p\left(\frac{1}{\sqrt{\prod_n N_n}}\right)$.

(iii). $\max_{i_1,\ldots,i_d}|X_{i_1,\ldots,i_d}|$ *bounded*

Assumption B.2.(i) is very similar to Assumption 4 except that here full rank is required after the within-cluster projection rather than the factor projection. Assumption B.2.(ii) is an exogeneity condition that requires weak exogeneity in the covariates after the within-cluster transformation, which can be viewed as similar to Assumption 2. This is stricter than Assumption 2 because the noise term ε can foreseeably impact cluster allocation if clusters are estimated as functionals of a residual term. This limitation is alleviated by, for instance, making sure cluster assignments are based on variables extraneous to the regression line, hence independent of ε. Likewise, independence can be achieved with a sample split, such as that proposed in Freeman and Weidner (2022). Here clusters could be formed from estimates made in one part of the split, and β estimated from the other part of the split.

Assumption B.2.(iii) requires that $X_{i_1,\ldots,i_d}$ has bounded support. This regularity condition is required for the proof strategy, but may not be strictly necessary. Simulations use normally distributed random variables to generate covariates, with good finite sample results. This assumption can be relaxed, but would require additional regularity on the space of fixed-effects, see discussion under Assumption 6.

Proposition B.1 (Upper bound on group fixed-effects estimator). *Let Assumptions B.1 and B.2 hold for cluster allocation $\mathcal{C}$. Let $\mathcal{M}$ be the set defined in Assumption B.1.(ii). Then, for tensor rank L that may depend on sample size,*

$$\|\widehat{\beta}_{GFE,\mathcal{C}} - \beta^0\| = \sqrt{L}O_p\left(\prod_{n^*\in\mathcal{M}}\sqrt{\xi_{N_{n^*}}}\right) + O_p\left(\prod_{n=1}^{d}\frac{1}{\sqrt{N_n}}\right).$$

Discussed in Section 4.2 are methods and conditions that restrict $\xi_{N_{n^*}}$ from Assumption B.1 and Proposition B.1 to $1/N_{n^*}$. This suggests that as long as $\mathcal{M} = \{1, \ldots, d\}$ and L_N is bounded the parametric rate of convergence is achievable. Related to $\mathcal{M}$, an implicit requirement on the latent parameters from this subset of dimensions is some form of low-dimensionality in the vectors $\varphi^{(n^*)}_{i_{n^*}}$ for $n^* \in \mathcal{M}$. For a discussion on the curse of dimensionality using clustering methods, see Bonhomme, Lamadon and Manresa (2022) that suggests the dimension of these parameters should be ≤ 2 for clustering to be a good approximation. This can, however, be alleviated if the within-cluster projection is performed iteratively one ℓ at a time, such that clusters are only ever formed over scalars $\varphi^{(n^*)}_{i_{n^*}\ell}$, rather than the full vectors, $\varphi^{(n^*)}_{i_{n^*}}$.

A sufficient condition for parameters in each dimension to be low-dimensional is low multilinear rank for each $n^* \in \mathcal{M}$. Hence, it is expected that the set $\mathcal{M}$ should be a subset of $\mathcal{L}$, from Assumption 3. The advantage of the group fixed-effects methods is that the analyst does not need to choose which n does admit low multilinear rank, hence it is more flexible.

Remark B.1. *In the three dimensional setting, the group fixed-effect objective function,*

$$Q(\beta, \mathbf{g}) = \min_{\theta_1,\theta_2,\theta_3} \sum_{i,j,t} \left(Y_{ijt} - X'_{ijt}\beta - \theta_{1;g_1(i)jt} - \theta_{2;ig_2(j)t} - \theta_{3;ijg_3(t)}\right)^2 \tag{B.2}$$

where $\theta_1 \in \mathbb{R}^{\mathrm{ncol}(g_1)\times N_2 \times N_3}$, $\theta_2 \in \mathbb{R}^{N_1\times\mathrm{ncol}(g_2)\times N_3}$, and $\theta_3 \in \mathbb{R}^{N_1\times N_2\times\mathrm{ncol}(g_3)}$; and $\mathrm{ncol}(\cdot)$ *returns the number of columns of a matrix. Then, taking the within-cluster transformation on $\mathbf{Y}$ and $\mathbf{X}$ from* (3) *is equivalent to differencing out the minimisers from* (B.2). *For example, $g_1(i)$ maps to the group identity of individual i. This is why θ_1 is restricted to vary across only* $\mathrm{ncol}(g_1)$ *different values in the first dimension.*

Notice that the optimisers for θ_1, θ_2 and θ_3 from (B.2) *can be described as combinations of the within-cluster projection as follows,*

$$\begin{aligned}
\widehat{\theta}_{1;g_1(i)jt} &\approx \bar{\mathcal{A}}_{i^*jt} - \bar{\mathcal{A}}_{i^*j^*t} + \bar{\mathcal{A}}_{i^*j^*t^*}\\
\widehat{\theta}_{2;ig_2(j)t} &\approx \bar{\mathcal{A}}_{ij^*t} - \bar{\mathcal{A}}_{i^*j^*t}\\
\widehat{\theta}_{2;ijg_3(t)} &\approx \bar{\mathcal{A}}_{ijt^*} - \bar{\mathcal{A}}_{i^*jt^*},
\end{aligned}$$

though this representation is not unique.

Additional to controlling for any additive terms, this projection leaves the following interactive fixed-effects residual,

$$\widetilde{\mathcal{A}}_{ijt} = \sum_{\ell=1}^{L} (\varphi^{(1)}_{i\ell} - \bar{\varphi}^{(1)}_{i^*\ell})(\varphi^{(2)}_{j\ell} - \bar{\varphi}^{(2)}_{j^*\ell})(\varphi^{(3)}_{t\ell} - \bar{\varphi}^{(3)}_{t^*\ell}). \tag{B.3}$$

where $\bar{\varphi}^{(1)}_{i^*\ell}$ is the group mean of $\varphi^{(1)}_{i^*\ell}$ for the i^*'s in i's group, and so on for the other terms. Hence, sufficient projection of the interactive fixed-effects terms relies on the weaker condition *that parameters converge to their group means, namely,* $\varphi_{i\ell}^{(1)} \to \bar{\varphi}_{i^*\ell}^{(1)}$, $\varphi_{j\ell}^{(2)} \to \bar{\varphi}_{j^*\ell}^{(2)}$ *or* $\varphi_{t\ell}^{(3)} \to \bar{\varphi}_{t^*\ell}^{(3)}$ *for each* ℓ*. Indeed, the group mean differencing could be seen as a weighted mean difference across the population, with equal weight given to observations within the cluster and zero weight to observations outside of each cluster. This fact is utilised for the more generic kernel weighted difference estimator in Section 3.2, which is synonymous to a Nadaraya-Watson type estimator for each fixed-effects term.*

So far it has been shown that with the relatively innocuous shift from the within transformation to the within-cluster transformation, any additive terms are automatically controlled for and there are conditions to also control for the interactive term. Choice of clusters for this transformation is key for consistency in this setting. Given a set of proxies to cluster on, clustering or matching methods can be used to find these groups, for example Bonhomme, Lamadon and Manresa (2022). Developing a set of proxies to cluster on is important and is discussed in Section 4.2.

B.2 Group fixed-effect convergence result

Before discussing the result from Proposition B.1, an alternative restriction on the cluster assignments is proposed. If clustering is performed on a proxy measure of the fixed-effect then Assumption B.1 can be stated in terms of the proxies, which forms the statement of Remark B.2. This requires that the proxies form an injective mapping to the true fixed-effect parameters. An example of this are the conditions imposed in Freeman and Weidner (2022), stated in similar terms here:

Remark B.2 (Clustering)**.** *The statement of Assumption B.1 can be reformulated in terms of the cluster proxies as follows. Let* $\widehat{\varphi}_{i_n}^{(n)} := \widehat{\varphi}^{(n)}\big(\varphi_{i_n}^{(n)}\big) \in \mathbb{R}^{\widehat{r}_n}$ *be the proxy for individual* i_n *used to cluster along dimension* n*. Then,*

(i). For all n *as* $N_n \to \infty$*,*

$$\frac{1}{N_n}\sum_{i_n}^{N_n}\left\|\widehat{\varphi}_{i_n}^{(n)} - \widehat{\varphi}_{j_n(i_n)}^{(n)}\right\|^2 \lesssim O_p(1)$$

(ii). For a non-empty subset $\mathcal{M} \subset \{1,\ldots,d\}$ *take for any* $n^* \in \mathcal{M}$ *a sequence* $\xi_{N_{n^*}} \to 0$ *as* $N_{n^*} \to \infty$*. Then,*

$$\frac{1}{N_{n^*}}\sum_{i_{n^*}}^{N_{n^*}}\left\|\widehat{\varphi}_{i_{n^*}}^{(n^*)} - \widehat{\varphi}_{j_{n^*}(i_{n^*})}^{(n^*)}\right\|^2 = O_p(\xi_{N_{n^*}})$$

(iii). Let $\varphi_{i_n}^{(n)} \in \Phi_n$ *be the* r_n*-column vector of fixed effects, where* Φ_n *are convex sets for each* n*. For each* $a, b \in \Phi_n$ *there exists a scalar* $c_n > 0$ *such that* $\|a-b\| \leq c_n \cdot \left\|\widehat{\varphi}^{(n)}(a) - \widehat{\varphi}^{(n)}(b)\right\|$

If these alternate restrictions hold along with Assumption B.2, then the bound in Proposition B.1 holds for the GFE estimator.

Restrictions (i) and (ii) in Remark B.2 are exactly Assumption B.1.(i) and (ii) but with cluster proxies in place of the true parameter values. These are high level restrictions on the clustering mechanism that requires the mechanism to find closeness in the proxy space. Restriction (iii) in Remark B.2 is an injectivity assumption on the proxy functions that demands closeness in the underlying parameter space given closeness in the proxy space. This requires that the proxies do actually provide a mapping to the true parameter space, that is, that they are reasonable proxies. An example of proxies that do this are the singular vectors from Section 4.2 that fit the requirements of Lemma 1. To see this expand the term $\|a-b\|$ and use the triangle inequality to see, $\|a-b\| \leq \|a-\widehat{\varphi}^{(n)}(a)\|+\|\widehat{\varphi}^{(n)}(b)-b\|+\|\widehat{\varphi}^{(n)}(a)-\widehat{\varphi}^{(n)}(b)\|$, where the first two terms are bound at the rate $O_p(C_n^{-1})$. Note the rotation matrices are ignored for brevity and C_n is the convergence rate from Lemma 1. Hence, asymptotically, Remark B.2.(iii) can be achieved with $c_n = 1$. Again, this argument requires knowledge of the true error term $\boldsymbol{\mathcal{V}} = \boldsymbol{\mathcal{A}} + \boldsymbol{\varepsilon}$, which may be unreasonable. By similar discussion from the kernel weighted estimator, this condition might relax to the mean squared deviation from Section 4.2, but is not formally discussed further here.

The distance of the proxies, $\|\widehat{\varphi}^{(n)}(a)-\widehat{\varphi}^{(n)}(b)\|$, is difficult to bound using clustering methods when the dimension of the proxies are larger than two, see Graf and Luschgy (2002) and further discussion in Bonhomme, Lamadon and Manresa (2022). This implies that a low-dimensional set of proxies must bound the true parameter values for clustering methods to work well in this setting. Hence, whilst the relationship in restriction (iii) of Remark B.2 may be satisfied for an arbitrarily high-dimensional set of proxies, for a reasonable family of cluster mechanisms to bound these proxies as per restrictions (i) and (ii) of this remark, restriction (iii) must also hold for a low-dimensional set of proxies. This can be highly restrictive with, for example, fixed-effects $\varphi_{i_n}^{(n)}$ that are high-dimensional. This can be alleviated by iterating the projection of fixed-effects. That is, project the first fixed-effect, recast the model with the first fixed-effect projected out, and repeat the projection exercise on the second fixed-effect, and so on. This simplifies the problem to only projecting a one-dimensional object at each iteration.

For the remainder of this discussion section, allow L_N to grow as a function of total sample size, $N := \prod_{n=1}^{d} N_n$. How the sequences $\xi_{N_{n^*}}$ converges to zero and how L_N is bounded are important for the convergence result in Proposition B.1. First note that for fixed L_N the first term in the result simplifies to $O_p\left(\prod_{n^*\in\mathcal{M}} \sqrt{\xi_{N_{n^*}}}\right)$. Also note, if the conditions for Lemma 1 hold and clustering is based on singular vector estimates that adhere to Remark B.2, then it is possible to achieve $\xi_{N_{n^*}} = O_p\left(\frac{1}{\min\{N_{n^*},\prod_{n\neq n^*} N_n\}}\right)$. If each N_n grow at the same rate then the consistency result is,

$$\left\|\widehat{\beta}_{GFE,\mathcal{C}}-\beta^0\right\|=\sqrt{L_N}O_p\left(N_n^{-|\mathcal{M}|/2}\right).$$

In the worst case scenario $\sqrt{L_N}$ is upper bound by $\sqrt{L_N}\lesssim N_n^{(d-1)/2}$, which is taken from $L_N\leq\min_n\prod_{n'\neq n}N_{n'}$. The convergence result is then $O_p\left(N_n^{(d-1-|\mathcal{M}|)/2}\right)$, which is of course conservative but shows that if $|\mathcal{M}|=d$, then consistency is guaranteed albeit at the slow rate of $N_n^{1/2}$. This means that all dimensions must have good cluster assignments, which is obviously not an ideal worst case but shows the limitations of this method when L_N is unrestricted.

For the special case of $d=3$ it can be shown that $L_N\leq\min_n\prod_{n'\neq n}r_{n'}$. From the discussion above, it is expected that $n\in\mathcal{M}$ is sufficient for $n\in\mathcal{L}$, that is, r_n is small for the set of dimensions $n\in\mathcal{M}$. This tightens the bound in Proposition B.1 to $O_p\left(N_n^{\max\{-|\mathcal{M}|/2,1-|\mathcal{M}|\}}\right)$, such that only $|\mathcal{M}|\geq 1$ is required for consistency. The analogous tensor rank bound is so far not known for the case with $d\geq 4$.

Proof of Proposition B.1. In the following, let $\operatorname{vec}_K(\widetilde{\mathbf{X}})$ be the $\prod_n N_n\times K$ matrix of vectorised covariates after the within-cluster transformations where each column is a vectorised transformed covariate. The vec operator on other variables is the standard vectorisation operator. Also let $N=\prod_n N_n$ and the subscript $i=1,\dots,N$ be the index for the vectorised data when i has no subscript. Then,

$$\begin{aligned}
\beta_{GFE,\mathcal{C}}&=\left(\operatorname{vec}_K(\widetilde{\mathbf{X}})'\operatorname{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)}\operatorname{vec}_K(\widetilde{\mathbf{X}})'\operatorname{vec}(\widetilde{\mathbf{Y}})\\
&=\left(\operatorname{vec}_K(\widetilde{\mathbf{X}})'\operatorname{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)}\operatorname{vec}_K(\widetilde{\mathbf{X}})'\left(\operatorname{vec}_K(\widetilde{\mathbf{X}})\beta^0+\operatorname{vec}(\widetilde{\boldsymbol{\mathcal{A}}})+\operatorname{vec}(\widetilde{\boldsymbol{\varepsilon}})\right)\\
&=\beta^0+\left(\operatorname{vec}_K(\widetilde{\mathbf{X}})'\operatorname{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)}\operatorname{vec}_K(\widetilde{\mathbf{X}})'\left(\operatorname{vec}(\widetilde{\boldsymbol{\mathcal{A}}})+\operatorname{vec}(\widetilde{\boldsymbol{\varepsilon}})\right),
\end{aligned}$$

such that,

$$\begin{aligned}
\left\|\beta_{GFE,\mathcal{C}}-\beta^0\right\|&=\left\|\left(\operatorname{vec}_K(\widetilde{\mathbf{X}})'\operatorname{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)}\operatorname{vec}_K(\widetilde{\mathbf{X}})'\left(\operatorname{vec}(\widetilde{\boldsymbol{\mathcal{A}}})+\operatorname{vec}(\widetilde{\boldsymbol{\varepsilon}})\right)\right\|\\
&\leq\|\kappa_N\|+\|\omega_N\|
\end{aligned}$$

where

$$\begin{aligned}
\|\kappa_N\|&:=\left\|\left(\operatorname{vec}_K(\widetilde{\mathbf{X}})'\operatorname{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)}\operatorname{vec}_K(\widetilde{\mathbf{X}})'\operatorname{vec}(\widetilde{\boldsymbol{\mathcal{A}}})\right\|;\\
\|\omega_N\|&:=\left\|\left(\operatorname{vec}_K(\widetilde{\mathbf{X}})'\operatorname{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)}\operatorname{vec}_K(\widetilde{\mathbf{X}})'\operatorname{vec}(\widetilde{\boldsymbol{\varepsilon}})\right\|.
\end{aligned}$$

The terms κ_N and ω_N are dealt with separately.

First to bound κ_N. Notice,

$$\|\kappa_N\|\leq\left\|\left(\operatorname{vec}_K(\widetilde{\mathbf{X}})'\operatorname{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)}\right\|\left\|\operatorname{vec}_K(\widetilde{\mathbf{X}})'\operatorname{vec}(\widetilde{\boldsymbol{\mathcal{A}}})\right\|$$

Focus on the right hand part, and let $\langle .,. \rangle_F$ be the Frobenius inner product,

$$\left\| \mathrm{vec}_K(\widetilde{\mathbf{X}})' \mathrm{vec}(\widetilde{\boldsymbol{\mathcal{A}}}) \right\| = \left\| \begin{bmatrix} \langle \widetilde{X}_1, \widetilde{\boldsymbol{\mathcal{A}}} \rangle_F \\ \vdots \\ \langle \widetilde{X}_K, \widetilde{\boldsymbol{\mathcal{A}}} \rangle_F \end{bmatrix} \right\| \leq \left\| \begin{bmatrix} \left| \sum_{i=1}^N \widetilde{X}_{i,1} \widetilde{\mathcal{A}}_i \right| \\ \vdots \\ \left| \sum_{i=1}^N \widetilde{X}_{i,K} \widetilde{\mathcal{A}}_i \right| \end{bmatrix} \right\| \tag{B.4}$$

where the triangle inequality is used entry-wise. Then,

$$\left| \sum_{i=1}^N \widetilde{X}_{i,k} \widetilde{\mathcal{A}}_i \right| \leq \sum_{i=1}^N \left| \widetilde{X}_{i,k} \right| \left| \widetilde{\mathcal{A}}_i \right| \leq \max_i \left(\widetilde{\mathbf{X}}_{k,i} \right) \left\| \mathrm{vec}(\widetilde{\boldsymbol{\mathcal{A}}}) \right\|_1 \quad \text{for each } k = 1, \ldots, K$$

From Assumption 7.(iii) there is $\max_i \left(\widetilde{\mathbf{X}}_{k,i} \right) = O_p(1)$. This leaves $\left\| \mathrm{vec}(\widetilde{\boldsymbol{\mathcal{A}}}) \right\|_1$. Take $g_n(i_n)$ as the indices in i_n's cluster such that $|g_n(i_n)|$ is the cluster size. Also, let $\bar{\varphi}^{(n)}_{i_n^*}$ be the cluster average for i_n's cluster. Then,

$$\begin{aligned}
\left\| \mathrm{vec}(\widetilde{\boldsymbol{\mathcal{A}}}) \right\|_1 &= \sum_{i_1,\ldots,i_d} \left| \sum_{\ell=1}^L \prod_{n=1}^d \left(\varphi^{(n)}_{i_n,\ell} - \bar{\varphi}^{(n)}_{i_n^*,\ell} \right) \right| \\
&= \sum_{i_1,\ldots,i_d} \left| \sum_{\ell=1}^L \prod_{n=1}^d \left(\varphi^{(n)}_{i_n,\ell} - \frac{1}{|g_n(i_n)|} \sum_{j_n \in g_n(i_n)} \varphi^{(n)}_{j_n,\ell} \right) \right| \\
&= \sum_{i_1,\ldots,i_d} \left| \sum_{\ell=1}^L \prod_{n=1}^d \frac{1}{|g_n(i_n)|} \sum_{j_n \in g_n(i_n)} \left(\varphi^{(n)}_{i_n,\ell} - \varphi^{(n)}_{j_n,\ell} \right) \right| \\
&\leq \prod_{n=1}^d \sum_{i_n} \frac{1}{|g_n(i_n)|} \sum_{j_n \in g_n(i_n)} \left\| \varphi^{(n)}_{i_n} - \varphi^{(n)}_{j_n} \right\|.
\end{aligned}$$

Take for each n,

$$\begin{aligned}
\sum_{i_n} \frac{1}{|g_n(i_n)|} \sum_{j_n \in g_n(i_n)} \left\| \varphi^{(n)}_{i_n} - \varphi^{(n)}_{j_n} \right\| &\leq N_n^{1/2} \left(\sum_{i_n} \left(\frac{1}{|g_n(i_n)|} \sum_{j_n \in g_n(i_n)} \left\| \varphi^{(n)}_{i_n} - \varphi^{(n)}_{j_n} \right\| \right)^2 \right)^{1/2} \\
&\leq N_n^{1/2} \left(\sum_{i_n} \max_{j_n \in g_n(i_n)} \left\| \varphi^{(n)}_{i_n} - \varphi^{(n)}_{j_n} \right\|^2 \right)^{1/2} \\
&= O_p\left(N_n \sqrt{\xi_{N_n}} \right)
\end{aligned}$$

where the last line is a consequence of Assumption B.1. Lastly, Assumption B.2.(i) implies the left hand term of $\|\kappa_N\|$ is $O_p(1/\prod_n N_n)$. This leaves

$$\|\kappa_N\| = O_p\left(\prod_{n^* \in \mathcal{M}} \sqrt{\xi_{N_{n^*}}} \right)$$

Finally, to bound $\|\omega_N\|$. Note that

$$\|\omega_N\| \leq \left\| \left(\text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)} \right\| \left\| \text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}(\widetilde{\boldsymbol{\varepsilon}}) \right\|.$$

Use Assumption 2 to bound the right hand term, $\left\| \text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}(\widetilde{\boldsymbol{\varepsilon}}) \right\| = O_p\left(\sqrt{\prod_n N_n}\right)$. Then, as above, the left hand term is $O_p(1/\prod_n N_n)$ such that

$$\|\omega_N\| = O_p\left(\frac{1}{\sqrt{\prod_n N_n}}\right).$$

■

Proof of Remark B.2. Begin from B.6 in the proof of Proposition B.1. The right hand terms, $\left\| \varphi_{i_n}^{(n)} - \varphi_{j_n}^{(n)} \right\|^2$, are bound as,

$$\left\| \varphi_{i_n}^{(n)} - \varphi_{j_n}^{(n)} \right\|^2 \leq c_n^2 \left\| \widehat{\varphi}_{i_n}^{(n)} - \widehat{\varphi}_{j_n}^{(n)} \right\|^2,$$

by Remark B.2.(iii). The result then follows immediately by applying the conditions Remark B.2.(i) and (ii) after this inequality. ■

Proposition B.2 (GFE convergence for finitely supported fixed-effects)**.**

Proof of Proposition B.2. In the following, let $\text{vec}_K(\widetilde{\mathbf{X}})$ be the $\prod_n N_n \times K$ matrix of vectorised covariates after the within-cluster transformations where each column is a vectorised transformed covariate. The vec operator on other variables is the standard vectorisation operator. Also let $N = \prod_n N_n$ and the subscript $i = 1, \ldots, N$ be the index for the vectorised data when i has no subscript. Then,

$$\begin{aligned}
\beta_{GFE,\mathcal{C}} &= \left(\text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)} \text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}(\widetilde{\mathbf{Y}}) \\
&= \left(\text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)} \text{vec}_K(\widetilde{\mathbf{X}})' \left(\text{vec}_K(\widetilde{\mathbf{X}})\beta^0 + \text{vec}(\widetilde{\boldsymbol{\mathcal{A}}}) + \text{vec}(\widetilde{\boldsymbol{\varepsilon}})\right) \\
&= \beta^0 + \left(\text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)} \text{vec}_K(\widetilde{\mathbf{X}})' \left(\text{vec}(\widetilde{\boldsymbol{\mathcal{A}}}) + \text{vec}(\widetilde{\boldsymbol{\varepsilon}})\right),
\end{aligned}$$

such that,

$$\begin{aligned}
\|\beta_{GFE,\mathcal{C}} - \beta^0\| &= \left\| \left(\text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)} \text{vec}_K(\widetilde{\mathbf{X}})' \left(\text{vec}(\widetilde{\boldsymbol{\mathcal{A}}}) + \text{vec}(\widetilde{\boldsymbol{\varepsilon}})\right) \right\| \\
&\leq \|\kappa_N\| + \|\omega_N\|
\end{aligned}$$

where

$$\begin{aligned}
\|\kappa_N\| &:= \left\| \left(\text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)} \text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}(\widetilde{\boldsymbol{\mathcal{A}}}) \right\|; \\
\|\omega_N\| &:= \left\| \left(\text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)} \text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}(\widetilde{\boldsymbol{\varepsilon}}) \right\|.
\end{aligned}$$

The terms κ_N and ω_N are dealt with separately.

First to bound κ_N. Notice,

$$\|\kappa_N\| \leq \left\| \left(\text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}_K(\widetilde{\mathbf{X}})\right)^{(-1)} \right\| \left\| \text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}(\widetilde{\boldsymbol{\mathcal{A}}}) \right\|$$

Focus on the right hand part, and let $\langle .,. \rangle_F$ be the Frobenius inner product,

$$\left\|\text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}(\widetilde{\boldsymbol{\mathcal{A}}})\right\| = \left\| \begin{bmatrix} \langle \widetilde{X}_1, \widetilde{\boldsymbol{\mathcal{A}}} \rangle_F \\ \vdots \\ \langle \widetilde{X}_K, \widetilde{\boldsymbol{\mathcal{A}}} \rangle_F \end{bmatrix} \right\| \leq \left\| \begin{bmatrix} \sum_{i=1}^N \left| \widetilde{X}_{i,1}\widetilde{\mathcal{A}}_i \right| \\ \vdots \\ \sum_{i=1}^N \left| \widetilde{X}_{i,K}\widetilde{\mathcal{A}}_i \right| \end{bmatrix} \right\| \tag{B.5}$$

where the triangle inequality is used entry-wise. By Hölder's inequality

$$\sum_{i=1}^N \left| \widetilde{X}_{i,k}\widetilde{\mathcal{A}}_i \right| \leq \left\|\text{vec}(\widetilde{\mathbf{X}}_k)\right\| \left\|\text{vec}(\widetilde{\boldsymbol{\mathcal{A}}})\right\| \quad \text{for each } k = 1, \dots, K$$

This bounds the norm in (B.5) as,

$$\left\|\text{vec}_K(\widetilde{\mathbf{X}})'\text{vec}(\widetilde{\boldsymbol{\mathcal{A}}})\right\| \leq \sqrt{\sum_{k=1}^K \left\|\text{vec}(\widetilde{\mathbf{X}}_k)\right\|^2} \left\|\text{vec}(\widetilde{\boldsymbol{\mathcal{A}}})\right\|.$$

From Assumption 1.(i) there is $\sqrt{\sum_{k=1}^K \left\|\text{vec}(\widetilde{\mathbf{X}}_k)\right\|^2} = O_p\left(\sqrt{\prod_n N_n}\right)$. This leaves $\left\|\text{vec}(\widetilde{\boldsymbol{\mathcal{A}}})\right\|$. Take $g_n(i_n)$ as the indices in i_n's cluster such that $|g_n(i_n)|$ is the cluster size. Also, let $\bar{\varphi}_{i_n^*}^{(n)}$ be the cluster average for i_n's cluster. Then,

$$\begin{aligned}
\left\|\text{vec}(\widetilde{\boldsymbol{\mathcal{A}}})\right\|^2 = \left\|\widetilde{\boldsymbol{\mathcal{A}}}\right\|_F^2 &= \sum_{i_1,\dots,i_d} \left(\sum_{\ell=1}^L \prod_{n=1}^d \left(\varphi_{i_n,\ell}^{(n)} - \bar{\varphi}_{i_n^*,\ell}^{(n)} \right) \right)^2 \\
\text{(Jensen's inequality)} \qquad &\leq L^2 \sum_{i_1,\dots,i_d} \sum_{\ell=1}^L \frac{1}{L} \prod_{n=1}^d \left(\varphi_{i_n,\ell}^{(n)} - \bar{\varphi}_{i_n^*,\ell}^{(n)} \right)^2 \\
&\leq L \left(\prod_{n=1}^d N_n \right) \prod_{n=1}^d \frac{1}{N_n} \sum_{i_n} \left\| \varphi_{i_n}^{(n)} - \bar{\varphi}_{i_n^*}^{(n)} \right\|^2.
\end{aligned}$$

Expand the term, $\left\| \varphi_{i_n}^{(n)} - \bar{\varphi}_{i_n^*}^{(n)} \right\|^2$,

$$\begin{aligned}
\left\| \varphi_{i_n}^{(n)} - \bar{\varphi}_{i_n^*}^{(n)} \right\|^2 &= \left\| \varphi_{i_n}^{(n)} - \frac{1}{|g_n(i_n)|} \sum_{j_n \in g_n(i_n)} \varphi_{j_n}^{(n)} \right\|^2 \\
&\leq \frac{1}{|g_n(i_n)|^2} \left(\sum_{j_n \in g_n(i_n)} \left\| \varphi_{i_n}^{(n)} - \varphi_{j_n}^{(n)} \right\| \right)^2 \\
&\leq \max_{j_n \in g_n(i_n)} \left\| \varphi_{i_n}^{(n)} - \varphi_{j_n}^{(n)} \right\|^2
\end{aligned} \tag{B.6}$$

Then by Assumption B.1,

$$\left\|\mathrm{vec}_K\big(\widetilde{\mathbf{X}}\big)'\mathrm{vec}\big(\widetilde{\boldsymbol{\mathcal{A}}}\big)\right\| \leq \sqrt{L}\left(\prod_{n=1}^{d} N_n\right) O_p\left(\prod_{n\in\mathcal{M}} \sqrt{\xi_{N_n}}\right)$$

Lastly, Assumption B.2.(i) implies the left hand term of $\|\kappa_N\|$ is $O_p(1/\prod_n N_n)$. This leaves

$$\|\kappa_N\| = \sqrt{L}O_p\left(\prod_{n^*\in\mathcal{M}} \sqrt{\xi_{N_{n^*}}}\right)$$

Finally, to bound $\|\omega_N\|$. Note that

$$\|\omega_N\| \leq \left\|\left(\mathrm{vec}_K\big(\widetilde{\mathbf{X}}\big)'\mathrm{vec}_K\big(\widetilde{\mathbf{X}}\big)\right)^{(-1)}\right\|\left\|\mathrm{vec}_K\big(\widetilde{\mathbf{X}}\big)'\mathrm{vec}\big(\widetilde{\boldsymbol{\varepsilon}}\big)\right\|.$$

Use Assumption 2 to bound the right hand term, $\left\|\mathrm{vec}_K\big(\widetilde{\mathbf{X}}\big)'\mathrm{vec}\big(\widetilde{\boldsymbol{\varepsilon}}\big)\right\| = O_p\left(\sqrt{\prod_n N_n}\right)$. Then, as above, the left hand term is $O_p(1/\prod_n N_n)$ such that

$$\|\omega_N\| = O_p\left(\frac{1}{\sqrt{\prod_n N_n}}\right).$$

■